

Tau Core: An Atemporal Morphological Readout Framework and a Conditional Minimum-Viable Physical Completion

Foundation Paper I

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Abstract

Tau Core asks whether observed four-dimensional physics can be represented as an observer-indexed, information-losing readout of a richer atemporal morphological object. We first define the minimal architecture $s \mapsto \rho_\tau(s) \mapsto M_\tau \mapsto \text{Desc}_O(M_\tau)$, together with its source-freeze, null-quotient, and no-double-counting conditions. We then go beyond a framework-only statement by fixing one minimum-viable completion at the level of a common variational packet. In that completion a stabilized body M_* is solved before terminal variation, and a single once-counted generating functional supplies coframe, gauge/current, quantum-effect, and clock rows. Under stated Lorentzian, regularity, state/effect, and observer-carrier assumptions, the construction has compatible conditional limits: a semi-classical Einstein equation with renormalized quantum stress and the fixed-background Schrödinger–Born description. The same frozen completion also yields a falsifiable observer-resolution inequality

$$\delta t_{\min,O} \geq \frac{2\hbar}{\Delta H} \arcsin \frac{\epsilon_O}{2} \geq \frac{\hbar \epsilon_O}{\Delta H},$$

where $x_A = \epsilon_O^2$ is an independently reconstructed dimensionless body-action record-separation ratio. In the adopted common-Gram minimum-lift completion, Uhlmann minimization proves $x_A = x_Q = 2(1 - \sqrt{F})$ and $D_B = 2 \arcsin(\epsilon_O/2)$, so ϵ_O lower-bounds the minimum Fubini–Study/Bures record angle. The equality $x_A = x_Q$, reconstructed by independent action-curvature and state-tomography protocols, is the primary Tau-specific discriminator. Neither the inverse-energy dependence nor the standard chord-to-angle conversion is claimed as new. The Tau-specific candidate is instead the source-frozen physical identification of the morphological Gram chord with the observer carrier’s Uhlmann-amplitude distance. Relative to the present source axioms, physical record-edge occupation and this finite body–Uhlmann identity are proved logically independent: neither selects the other, although a realized test requires both together with an operational hold condition. For finite source algebras, we also derive an exact selection criterion: the multiplication/involution data, a faithful source state, and all source-owned distinguished roles determine one minimal cyclic completion up to typed unitary equivalence. Complete sector marginals do not determine this joint packet; an explicit faithful two-sector countermodel isolates the missing cross-sector source moments. The inequality concerns one stable record, not a universal discrete time atom. The result is a conditional theory closure: the current base–seed reduct admits alternative positive completions, and nature-level selection, fundamental-constant prediction, ultraviolet and Standard-Model completion, and empirical confirmation remain open. The manuscript therefore presents one auditable candidate with standard-limit checks, a hard claim ledger, and a predeclared discriminator, not a replacement for general relativity or quantum theory.

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1 Purpose and claim boundary

This manuscript states the Tau Core foundation language and the strongest currently frozen physical completion in a form that can be criticized, reproduced, and rejected. It is self-contained: no result from another Tau Core manuscript, repository, or unpublished analysis is used as a premise or as evidence for any claim made here.

The claim boundary has two levels. At the foundational level, the paper defines an atemporal body/readout architecture and its audit discipline. At the completion level, it fixes one common functional and proves consequences conditional on an explicit assumption ledger. Neither level supplies empirical validation or a theorem that the physical base-seed source selects that completion. The main exclusions are summarized below and itemized again near the end of the manuscript.

Reviewer-facing claim box. This paper claims (i) a formal body/readout architecture and failure discipline, and (ii) one *conditional* minimum-viable completion with compatible GR/optics and quantum/Born limits plus one frozen observer-resolution lower-bound discriminator. It does not claim empirical validation, uniqueness, nature-level source selection, a new force, dark-matter or dark-energy replacement, ultraviolet quantum gravity, or full Standard-Model recovery. Every conditional result below is void as a claim about Nature unless the common completion is independently selected or its predeclared discriminator survives testing.

The central idea can be written in one line:

$$s \mapsto \rho_\tau(s) \mapsto U_i^{M_\tau}(\rho_\tau(s)). \quad (1)$$

The first arrow is not time evolution. It is a parent-side response relation. The second arrow is a body-conditioned readout. The “readout” is not an extra physical substance or history store: in the completion developed below it is an observer-specific realization of ordering and compatibility already resident in the stabilized body. A physical branch appears only after the descent has supplied a codomain, a null policy, a closure test, and a validation boundary. When M_τ is fixed by the declared source class, the superscript is often suppressed and U_i is written for $U_i^{M_\tau}$.

In a deterministic branch one may write

$$\rho_\tau : \mathcal{S} \rightarrow \mathfrak{R}_\tau.$$

In a relational branch the same symbol denotes either a relation

$$\rho_\tau \subseteq \mathcal{S} \times \mathfrak{R}_\tau$$

or, equivalently, a set-valued response rule

$$\rho_\tau : \mathcal{S} \rightarrow \mathcal{P}(\mathfrak{R}_\tau).$$

Such a branch must declare whether it uses the whole admissible response class or an endpoint-blind source-freeze rule selecting a representative.

Equivalently, the paper works with the following foundation package:

$$\mathfrak{T} = (\mathcal{S}, \mathfrak{R}_\tau, \rho_\tau, M_\tau, \{U_i\}, \{N_i\}, \{J_i\}).$$

Here $\mathcal{S}$ is a seed/source domain, $\mathfrak{R}_\tau$ is a formal response domain, ρ_τ is a map, relation, or set-valued response rule, M_τ is the morphological body, U_i are sector readouts, N_i are null or quotient policies, and J_i are optional relifts. This package is an organizing object for later branches (Fig. 1), not a proof that Nature uses it.

In informal Tau Core discussions, the phrase *Tau Core base field* refers to this parent-side response substrate, represented here by $(\mathcal{S}, \mathfrak{R}_\tau, \rho_\tau)$, together with the morphological body M_τ . The phrase is heuristic. In this paper it does not denote a physical field with an action, equations of motion, units, stress-energy, or empirical calibration.

A. Minimal Tau Core spine

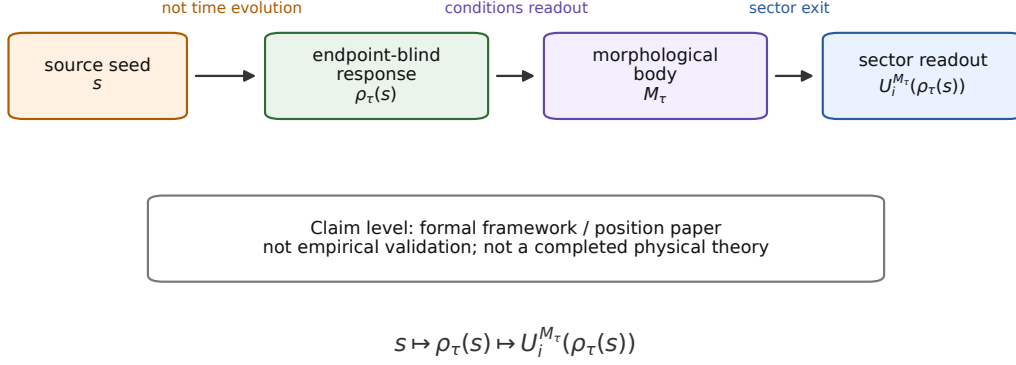


Figure 1: The minimal Tau Core spine used in Foundation Paper I. A parent-side seed s induces an endpoint-blind Tau response $\rho_\tau(s)$. A Tau-side morphological body M_τ conditions sector readouts U_i . The diagram is the weak foundation spine; the stronger conditional completion is shown in Fig. 4.

Method note. The manuscript was developed in an AI-assisted theoretical-research workflow. The AI workflow is not used as evidence for any physics claim; it is only part of the drafting and audit process.

This paper is therefore neither a phenomenology paper nor only a position paper. It is a foundation paper containing a conditional worked completion. It is useful only if that completion makes the program more falsifiable. A broad ontology that cannot be turned into frozen source rules, null tests, or endpoint-blind predictions is not useful as physics.

2 Reader roadmap

The paper has three jobs. First, it fixes the body/readout vocabulary. Second, it gives one explicit conditional completion and its proof dependencies. Third, it states exactly which physical-selection and empirical claims remain open.

Part of the paper	Role	Claim level
Opening sections	Motivation, name, and relation to block-universe and spacetime-field views	Positioning
Core vocabulary	Parent response, morphological body, observer descent, relift, and defect	Definitions
Conditional MVP	Body-first stabilization, one common functional, GR/optics and quantum/Born limits, QOR discriminator	Conditional results
Formal safeguards	Quotients, source discipline, effective readout-time, postulates, and finite examples	Lemmas and toy mechanisms
Boundary layer	Gravity, quantum, cosmology, adjacent theories, objections, and failure conditions	Interpretation and non-claims
Audit layer	Assumption/result ledger, dimensional checks, maturity checklist, and compact summary	Reviewer audit

Thus the paper should be read from weak to strong claim levels. Definitions do not become assumptions; conditional theorems do not establish physical selection; standard-limit recovery does not establish novelty; and a frozen prediction does not become empirical validation until it survives the declared test.

3 Why a foundation paper is needed

Several themes in modern foundations already question whether time, spacetime, or local dynamics should be primitive. Canonical quantum gravity contains the well-known problem of time, where a timeless constraint equation can coexist with effective dynamics [13, 34]. Relational time constructions show how dynamics can be described through correlations rather than external evolution [30]. Timeless or configuration-space views have also been explored philosophically and mathematically [4]. The thermal-time hypothesis is especially close in spirit because it treats time as state-dependent rather than as an external primitive [10]. Causal-set and generalized quantum-mechanics programs provide other examples of frameworks that change the primitive descriptive layer rather than merely adding a term to an equation [6, 19]. Quantum reconstruction programs show that large parts of quantum formalism can be organized by operational principles rather than simply postulated [18, 9, 27]. Constructor theory is a useful methodological comparison because it also tries to state physical content through possible/impossible transformations rather than through one specific dynamical equation [11, 12].

Tau Core does not claim to solve those programs. It uses a different organizing question:

Can observed sector descriptions be treated as readouts of a deeper parent response, so that time evolution, quantum state, gravity, mass, and observer-time are effective sector descriptions rather than primitive categories?

This question is intentionally broader than any one branch. If it is too broad, it becomes metaphysics. The purpose of the present paper is to prevent that drift by specifying the minimal objects that must exist before a Tau Core statement becomes mathematically meaningful.

The immediate motivation is the following tension. Many physical descriptions use variables that are defined only after an observer, a section, a clock, a measurement basis, or a spacetime

background is in place. But the same description often treats those variables as if they were primitive. Tau Core instead separates:

$$\text{parent response} \quad \text{from} \quad \text{sector readout}.$$

That separation is the entire point of the core-spine relation (1).

4 The name Tau Core

The name “Tau Core” is historical. The program began from a time-based intuition: perhaps the time seen by a four-dimensional observer is not a primitive background, but a projected response. This explains the use of τ , a symbol often associated with time or proper time in physics.

In the present architecture, however, τ does not mean ordinary time. It names the parent-state side of the theory. Time is one possible readout from that side. The distinction is important:

Symbol or phrase	Meaning here	Not claimed here
τ	parent-state side	ordinary time itself
X_τ	parent Tau state	detected physical field
$\rho_\tau(s)$	endpoint-blind response	time evolution
U_{time}	observer-indexed clock descent	nature-selected universal clock
M_τ	morphological body	observed 4D shape

Thus the theory is time-motivated but not time-reductionist. It does not say that everything is made of time. In the completion used here, causal orientation and ordering belong to the body, while measurable elapsed time and its observer resolution arise in the observer descent. The symbol τ names the deeper side; it is not an additional parent clock.

5 Difference from a block universe

A block-universe reading treats a four-dimensional spacetime history as the final object. The comparison target is not a straw man. The relativistic background begins with Minkowski’s four-dimensional spacetime formulation [29]. The Rietdijk–Putnam–Maxwell line of argument then uses relativity of simultaneity to motivate four-dimensionalism or eternalism [33, 32, 28, 39]. Standard surveys of the philosophy of spacetime and time describe the resulting debates around being, becoming, eternalism, and the block universe [35, 26, 31, 38]. In that setting one may summarize the block-universe reading as

$$\text{Reality} = M_{4D}. \tag{2}$$

Tau Core instead treats a four-dimensional block, when such a block is a good description, as a readout:

$$M_{4D} = U_{4D}^{M_\tau}(\rho_\tau(s)). \tag{3}$$

This difference has physical content only if additional non-spacetime readout constraints can be source-frozen beyond the spacetime block itself. If no non-spacetime readout imposes such restrictions, the distinction remains a metaphysical rephrasing rather than a physical branch. The difference is not cosmetic. In a block-universe view the four-dimensional object is the final arena. In Tau Core it is an arena produced by a readout. The parent response can in principle have distinctions that one readout identifies, another ignores, and a third translates into effective readout-variation.

B. Block-universe ontology versus Tau Core readout

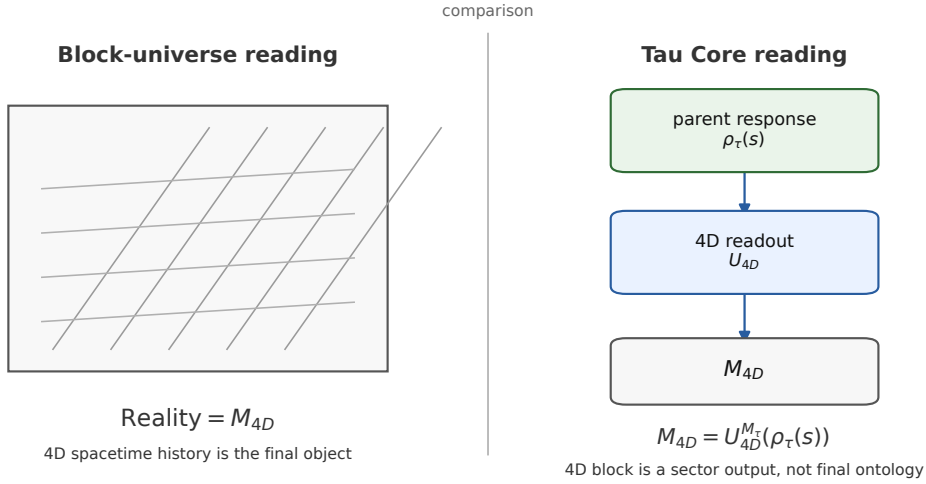


Figure 2: The block-universe comparison used in this paper. The left side represents the standard four-dimensional block reading, in which the spacetime history is the final object. The right side represents the Tau Core position: the same effective 4D block may be a stable readout $M_{4D} = U_{4D}^{M_\tau}(\rho_\tau(s))$ of a deeper parent response. The figure is a conceptual distinction, not a proof against eternalism.

Tau Core can be compatible with an effective 4D block without taking it as ontologically final or refuting block-universe accounts. It instead relocates the block from final ontology to sector output (Fig. 2):

$$\text{block as final object} \longrightarrow \text{block as } U_{4D}\text{-readout.}$$

6 Difference from ordinary spacetime field theory

An ordinary field theory begins with a field on spacetime:

$$\phi : M_{4D} \rightarrow V, \tag{4}$$

and then studies the dynamics of $\phi(x, t)$. Tau Core rewrites the order:

$$\phi_{\text{obs}}(x, t) = U_\phi(\rho_\tau(s)). \tag{5}$$

The observed field is a readout. The spacetime coordinates are also part of the readout structure. This does not make ordinary field theory wrong. It makes it effective: a very successful description after a particular sector readout has become stable.

This is why Tau Core cannot claim physical success merely by writing a new readout map. To become physics, a readout must recover known limits and must survive empirical controls. If U_ϕ is arbitrary, the framework is empty. The research program therefore needs source-frozen readout rules, quotient-safe null policies, and explicit failure conditions.

The MVP completion does not discard spacetime field theory. It recovers an ordinary Lorentzian effective field theory on a selected descended leaf and uses its standard variational machinery. The proposed difference lies upstream: the coframe, state/effect pair, and observer clock are jointly typed outputs of a frozen body rather than mutually independent primitives. Whether that upstream organization is physically real is exactly the selection and QOR-test question left open here.

7 Core vocabulary

Definition 1 (Parent seed). *A parent seed $s \in \mathcal{S}$ is a Tau-side source datum. It is not an observed endpoint and must not be chosen by looking at the endpoint residual that a later readout is supposed to explain.*

Definition 2 (Endpoint-blind Tau response). *An endpoint-blind Tau response is a map or relation*

$$\rho_\tau : \mathcal{S} \rightarrow \mathfrak{R}_\tau$$

whose input is the parent seed and admissible parent-side constraints, not the observational endpoint. Endpoint blindness is a non-leakage rule, not a proof that the response is physically real. When ρ_τ is relational rather than single-valued, $\rho_\tau(s)$ denotes the admissible response class, or a representative chosen by an explicitly declared source-freeze rule. No endpoint residual may be used to select the representative.

Definition 3 (Morphological body). *The Tau-side morphological body M_τ is the intrinsic structured body that conditions readouts. It is not an observed 4D morphology label. It may carry sector, branch, wall, null, Hessian-support, admissibility, global-order, and observer/path-relevant structure. A projected morphology is only a readout handle on this deeper body.*

For Foundation Paper I the minimal mathematical type is:

$$M_\tau \in \mathcal{M}_\tau^{\text{adm}},$$

where $\mathcal{M}_\tau^{\text{adm}}$ is a declared class of admissible morphological records. The required part of such a record is deliberately small: it must specify the source domain, the readout family, the null policy, and the admissibility/freeze rule used by the branch. Sector decompositions, walls, Hessian support, global order, and observer/path structure are optional branch data unless the branch explicitly uses them in its readout rule. A branch may instantiate such a record schematically as

$$M_\tau = (S, B, W, N, H, A; \Theta),$$

where S records sector decomposition, B branch/support structure, W wall or transition adjacency, N null/protected structure, H Hessian-response support, A admissibility constraints, and Θ optional interface/order/path data. Thus $U_i^{M_\tau}$ is not merely a decorative superscript: it means that the readout map is selected from a typed family

$$U_i : \mathcal{M}_\tau^{\text{adm}} \times \mathfrak{R}_\tau \rightarrow \mathcal{O}_i.$$

This is still a minimal type declaration, not a construction theorem for $\mathcal{M}_\tau^{\text{adm}}$. In particular, an observed galaxy morphology, a quantum access structure, and an observer-time structure may be different projected handles on different subrecords of M_τ , not identical kinds of morphology.

Definition 4 (Readout). *A readout is a morphology-conditioned sector map. Equivalently, one may write*

$$U_i^{M_\tau} : \mathfrak{R}_\tau \rightarrow \mathcal{O}_i, \quad \text{or} \quad U_i : \mathcal{M}_\tau \times \mathfrak{R}_\tau \rightarrow \mathcal{O}_i.$$

The codomain $\mathcal{O}_i$ may be a spacetime description, a time ordering, a quantum access state, a gravitational response, a mass/source record, or a more specialized observable structure. Each readout has its own null directions and validation boundary. When M_τ is fixed by the declared source class, we suppress it and write $U_i(r)$.

Readout family	Parent/source class	Readout form	Minimal freeze target
Spacetime	local chart and metric handles	$U_{4D}^{M\tau}(r)$	local interval or chart limit
Observer-time	clock, interface, and path records	$U_{\text{time}}^{M\tau}(r, \mathcal{O}, \gamma)$	proper-time limit or frozen path correction
Quantum access	access, phase, context, or outcome records	$U_q^{M\tau}(r)$	normalized access probabilities or a null/control rule
Gravity	weak-field, closure, lensing, or acceleration records	$U_g^{M\tau}(r)$	Newtonian/GR limit or quotient-visible residual
Mass/source	density, source-measure, flavor, or particle-content records	$U_m^{M\tau}(r)$	conserved source measure or declared mass/source map
Cosmology	web, void, path, clock, or expansion records	$U_{\text{cosmo}}^{M\tau}(r)$	FLRW/ Λ CDM limit or frozen residual descriptor

Table 1: Readout families and illustrative parent/source classes. The table does not derive the families physically; it records the minimal bookkeeping needed before a branch can become more than a vocabulary. The freeze targets are intentionally heterogeneous placeholders. Each later branch must replace them by operationally defined quantities with units, uncertainties, null models, and failure rules; they are not yet calibrated observables.

Definition 5 (Cross-scale readout composition). *Readout families are branch-distinct but not assumed independent. A candidate kernel K_i for one family is admissible only after the source, carrier, quotient, and observable dependencies that compositionally link it to neighboring readout families have been checked, or explicitly frozen as negligible under the declared assumptions. Equivalently, Tau Core treats physical readouts as a compatibility-coupled composition hypergraph, not as isolated fit modules.*

This is an admissibility rule, not a derivation of unification. It forbids treating a gravity kernel as independent of the particle, mass, material, radiation, and path readouts composing its carrier (Fig. 3). It also prevents double counting: two source-overlapping readout terms cannot be added as independent corrections unless their non-overlap and quotient visibility have been declared.

Definition 6 (Relift). *A relift for a readout U_i is a map or partial map*

$$J_i : \mathcal{O}_i \rightarrow \mathfrak{R}_\tau$$

that attempts to reconstruct a parent-response representative from a readout record. It need not exist globally and need not be unique.

Definition 7 (Readout obstruction). *Given $r = \rho_\tau(s)$, the primary defect object is not a subtraction but an obstruction record*

$$\omega_i(r) = 0 \iff J_i U_i(r) N_i r,$$

whenever a relift and null relation have been declared. Thus $\omega_i(r) \neq 0$ records that the readout and its allowed relift fail to recover the parent-response representative modulo the declared null relation. Only in a chosen affine or linearized chart may this be represented by the coordinate expression

$$h_i(r) = J_i(U_i(r)) - r.$$

The obstruction condition and its chart-level representative are not dark-matter formulas or quantum-collapse formulas. They are abstract incompleteness records. A physical branch must later specify the codomain, quotient, units, known limits, and empirical test.

D. Cross-scale readout composition

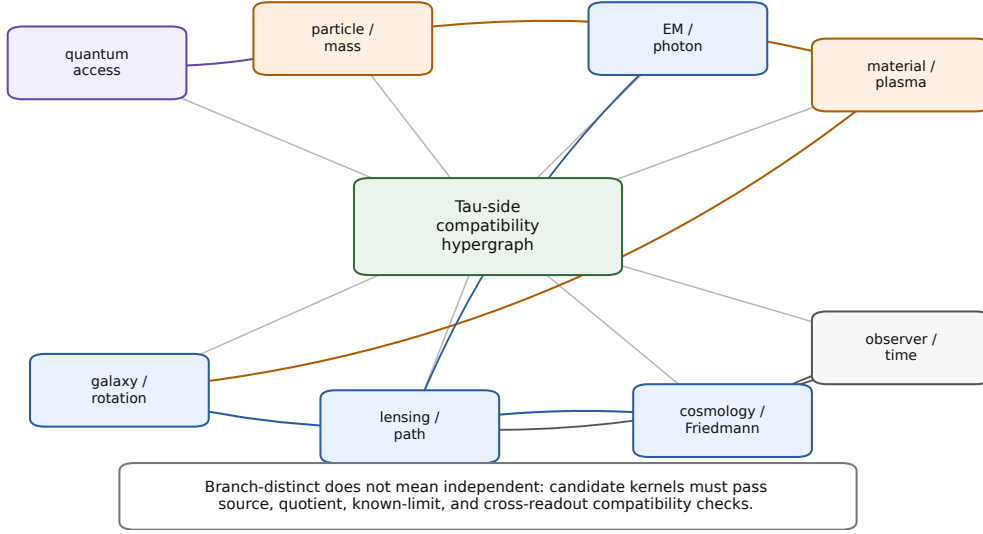


Figure 3: Cross-scale readout composition. Tau Core readout families are branch-distinct, but branch distinction does not imply operational independence. Candidate kernels must respect source, carrier, quotient, and cross-readout compatibility constraints before they can be promoted beyond scaffold status. The figure is architectural: it is not a proof of physical unification.

8 Status of the primitive objects

A common failure mode for foundation papers is to let definitions, assumptions, and derived claims blur into each other. The following ledger records the status of the new objects used above. It is not additional evidence; it is a claim-boundary device.

Primitive-object status ledger

Object	Status in the weak spine	Additional MVP status
$\mathcal{S}$	declared source/seed domain	one universal-seed branch is adopted, not selected by Nature
$\mathfrak{R}_\tau$	parent-response space, assumed as formal domain	one enriched variational completion is fixed
ρ_τ	endpoint-blind response map or relation	represented by a body action plus ordered descent
M_τ	morphology-conditioning body	a stationary M_* with positive quotient Hessian is assumed
$U_i^{M_\tau}$	sector readout definition	gravity/optics, quantum/Born, and QOR leaves are constructed conditionally
N_i	declared null/quotient policy	only common terminal nulls may be removed
J_i	optional relift device	not required as a global inverse in the MVP
ω_i	obstruction record modulo N_i	remains diagnostic, not a dark-sector or collapse mechanism

Section 10 replaces part of the former third-column gap with one conditional worked completion. The decisive remaining gap is no longer the absence of any common action or standard

limit; it is nature-level selection of that particular completion and empirical testing of its non-standard discriminator.

9 Minimal readout calculus

The formalism above is intentionally weak. A readout is not yet a functor, a field equation, a probability rule, or a variational principle. Nevertheless, some minimal calculus is needed if the framework is to avoid being only language.

The mathematics in this section is standard. A readout induces a partition of the parent-response space; quotient safety is well-definedness on the corresponding quotient; refinement is the ordinary refinement order on partitions. Tau Core does not claim novelty for these facts. It uses them as bookkeeping discipline so that later physical branches cannot hide endpoint leakage or double counting behind new terminology. The same conceptual territory overlaps with coarse-graining, marginalization, consistent-history and decoherence-inspired languages, and Jaynes-style information-theoretic statistical mechanics [17, 14, 23, 24].

Let $\mathfrak{R}_\tau$ be a parent-response space and let U_i be a readout. The first mathematical object induced by a readout is its indistinguishability relation:

$$r \sim_i r' \iff U_i(r) = U_i(r'). \quad (6)$$

This relation is readout-relative. A distinction can be invisible to one sector and visible to another. In this sense, a readout is not just a map to observables; it is also a rule for what counts as the same state for that sector.

Definition 8 (Readout kernel). *The readout kernel of U_i is the equivalence relation $\sim_i$ defined by the readout-equivalence relation (6). If $\mathfrak{R}_\tau$ has a linear structure and U_i is linear, this reduces to the usual null space $\text{Ker } U_i$. In general it is an equivalence relation, not necessarily a linear subspace.*

Definition 9 (Readout refinement). *A readout U_j refines U_i if*

$$r \sim_j r' \implies r \sim_i r'.$$

Equivalently, U_j distinguishes at least as much parent-response information as U_i .

This gives a minimal partial order on readouts. A more informative readout is not automatically more physical. It may be harder to stabilize, harder to observe, or more vulnerable to noise. But the refinement relation prevents one from treating all readouts as interchangeable.

Definition 10 (Joint readout). *Given readouts U_i and U_j , the joint readout is*

$$U_{ij}(r) = (U_i(r), U_j(r)).$$

Its equivalence relation is the intersection:

$$\sim_{ij} = \sim_i \cap \sim_j.$$

Proposition 1 (Joint readouts refine their components). *The joint readout U_{ij} refines both U_i and U_j .*

Proof. If $r \sim_{ij} r'$, then $U_i(r) = U_i(r')$ and $U_j(r) = U_j(r')$. Hence $r \sim_i r'$ and $r \sim_j r'$. Therefore the joint equivalence relation is contained in each component equivalence relation. $\square$

This trivial observation is one reason multi-readout tests are valuable, but it should not be confused with statistical independence. Two readouts may be correlated sibling readouts of the same parent. A minimal observation model is

$$R_i = O_i[P] + N_i,$$

where P is a shared latent parent, O_i is the sector readout operator, and N_i collects noise, selection, and pipeline effects. In cosmology, for example, galaxy clustering, weak lensing, gas tracers, and CMB lensing can all respond to the same matter/potential parent while using different observational operators.

Therefore raw agreement between readouts is not, by itself, a Tau Core signal. The stronger question is whether a common residual remains after conditioning on the appropriate null parent P_0 :

$$\Delta R_i = R_i - O_i[P_0].$$

For a cosmological branch P_0 might be a matched Λ CDM matter/potential parent; for another branch it would be the relevant standard baseline. A Tau Core-compatible excess would be residual cross-readout structure that survives the declared null model, covariance, and source-overlap ledger. Here this is simply called source-overlap discipline. It is not a validation statistic and not an independence claim. Without this discipline, a joint readout may simply double count the same source coordinate.

10 A conditional minimum-viable physical completion

The preceding architecture is deliberately weak: by itself it does not select a physical action, metric signature, quantum state, or observer. This section fixes one minimum-viable physical (MVP) object so that the paper contains an auditable theory chain rather than an indefinitely extensible list of possible readouts. The frozen chain includes a quantum observer-resolution (QOR) discriminator and is

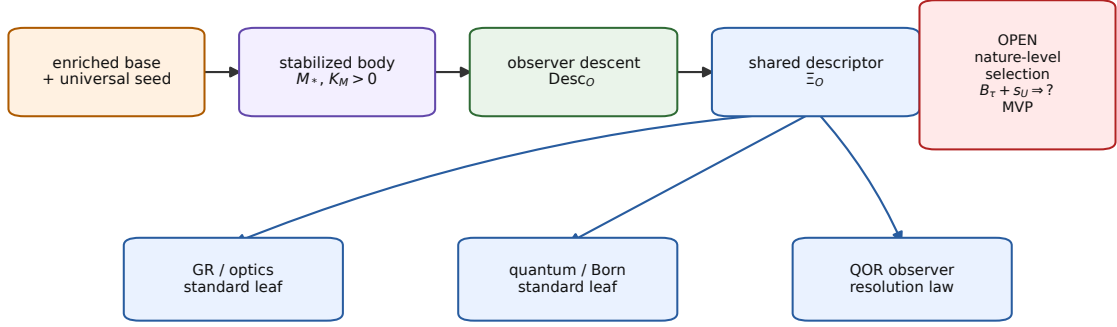
$$\boxed{\begin{array}{l} \Gamma_{\text{MVP}} \longrightarrow M_* \longrightarrow \text{Desc}_O(M_*) \\ \longrightarrow \{\text{GR/optics, quantum/Born, QOR}\}. \end{array}} \quad (7)$$

Every arrow in Eq. (7) is conditional. The chain is internally composable and has a frozen discriminator, but upstream selection in Nature remains open (Fig. 4).

10.1 Frozen assumption ledger

The completion uses the following assumptions. They are deliberately separated from consequences.

I. Conditional minimum-viable theory chain



Closed as one conditional completion; not selected or validated as Nature's completion.

Figure 4: The conditional minimum-viable chain. A physically enriched base and universal seed are assumed to stabilize a body before observer descent. One shared descriptor then supplies the standard leaves and the QOR discriminator. The red box is the exact open step: nature-level selection of this completion from the upstream base–seed source.

Label	Assumption used in the MVP completion
A1	A physically enriched base and one universal seed admit a stabilized body M_* with a twice-differentiable body action.
A2	After quotienting only directions annihilated by every claimed terminal, the body Hessian K_M is positive on the retained tangent space.
A3	The body is solved first and frozen; terminal variations do not re-stabilize it or feed endpoint information into its definition.
A4	Coframe, connection/gauge, occupied matter, clock, and effect probes are once-counted terms or sources of one post-body generating functional.
A5	The relevant descended leaf is orientable and Lorentzian; its gravitational term lies in the torsion-free Palatini/Einstein–Hilbert infrared class and is sufficiently regular for the variations used below.
A6	The quantum leaf admits positive normalized states and a complete effect family; the valuation on complete effects is additive and noncontextual.
A7	The field-theory leaf admits a locally covariant Hadamard prescription for the renormalized stress tensor and its local curvature ambiguities.
A8a-OCC	The physical completion occupies a nonzero source-owned recovery support, at least one record incidence, and a CP record instrument on that support. The finite pointer algebra and common Uhlmann amplitude bundle are then derived, not separately assumed.
A8b	The same post-body source that owns $x_O = \mathcal{A}_X^{PC}/\mathcal{A}_*$ selects a normalized amplitude connector whose horizontal Hilbert–Schmidt pullback equals the normalized body-action metric. On the explicit mixed-state completion it selects $V_{\text{phys}} = V_{r(x_O)}$, with $r(x_O)^2 = x_O - x_O^2/4$ and $0 < x_O \leq 2$.
A9	The quantum leaf uses the common action unit $\mathcal{A}_* = \hbar$, and self-adjoint transport obeys the Fubini–Study/Bures quantum-speed-limit bound.

A1–A4 define the working completion. A5–A9, including A8a-OCC and A8b, are standard-limit and prediction assumptions. None is claimed to follow from the old minimal Tau skeleton alone. In particular, positivity of a body Hessian does not select the Lorentzian leaf, the Born valuation, or a physical observer carrier. Section 10.9 proves a sufficient algebraic source theorem for A8b: affiliation of the global Dirac operator with the recovery relation algebra. The

displayed parent functional has not been shown to satisfy that law, so A8b remains a physical completion assumption.

10.2 Body-first variational order

Let $\mathcal{A}_{\text{body}}$ denote the atemporal body action. Stabilization means

$$D_M \mathcal{A}_{\text{body}}(M_*) = 0, \quad K_M = D_M^2 \mathcal{A}_{\text{body}}|_{M_*} > 0 \quad \text{on } T_{M_*} \mathcal{M} / \mathcal{N}_{\text{true}}. \quad (8)$$

Here $\mathcal{N}_{\text{true}}$ contains only directions annihilated by every terminal retained in the MVP. A direction invisible to one terminal is not discarded if another terminal sees it.

The order in Eq. (8) matters. If z denotes all post-body terminal variables, linear response is solved in the ordered form

$$K_M \delta M = -j_M, \quad (9)$$

$$K_z \delta z = -j_z - C \delta M, \quad (10)$$

where $C = D_z D_M \Gamma|_{M_*}$ is an allowed body-to-terminal coupling. Equations (10) define a lower-triangular response problem. Placing M and z in an unconstrained simultaneous symmetric fit would reverse the generative direction and permit endpoint data to alter the body. The completion therefore forbids that operation unless a separately sourced post-body copy of the body variable is introduced and justified.

10.3 One common post-body functional

Three differently typed objects must be kept separate: the pre-body action used to solve M_* , the post-body microscopic functional before quantum or apparatus variables are integrated out, and the renormalized observer-indexed effective functional afterwards. Their lawful order is

$$\mathcal{A}_{\text{pre}}[M; s, B] \xrightarrow{\text{stat}_M} M_*, \quad \Gamma_{\text{micro}, \text{O}}[\Phi; M_*, e, A, \eta], \quad \Gamma_{\text{MVP}} \equiv \Gamma_{\text{eff}, \text{O}}[M_*; e, A, \eta]. \quad (11)$$

The transferable body sector used in the first solve is not appended again downstream. An admissible once-counted microscopic ownership ledger is

$$\begin{aligned} \Gamma_{\text{micro}, \text{O}} = & S_{\text{grav}}^{\text{bare}}[e] + S_{\text{body}}^{\text{ret}}[M_*; e, \Phi_{\text{ret}}] \\ & + S_q^{\text{bare}}[M_*; e, A, \Phi_q] + S_{\text{src}, \text{O}}[M_*; e, A, \Phi_q, \Phi_{\text{ret}}; \eta, \vartheta], \end{aligned} \quad (12)$$

where every primitive monomial or incidence has exactly one owner. Integrating the declared quantum and apparatus variables once defines

$$e^{iW_O^{\text{ren}}[M_*; e, A, \Phi_{\text{ret}}, \eta, \vartheta]/\hbar} = \int_{\mathcal{C}_O} \mathcal{D}\Phi_q e^{i(S_q^{\text{bare}} + S_{\text{src}, \text{O}})/\hbar} \Big|_{\text{ren}} \quad (13)$$

and the effective packet

$$\boxed{\Gamma_{\text{MVP}} = S_{\text{EH}}[e] + S_{\text{loc}}^{\text{ren}}[e] + S_{\text{body}}^{\text{ret}}[M_*; e, \Phi_{\text{ret}}] + W_O^{\text{ren}}[M_*; e, A, \Phi_{\text{ret}}, \eta, \vartheta]}. \quad (14)$$

Here $S_{\text{body}}^{\text{ret}}$ is the retained post-body representative; its role is to preserve the occupied body value and first variation without copying the already used pre-body envelope. If observer degrees of freedom and their source insertions are already included in W_O^{ren} , there is no additional Γ_{obs} summand at the effective level. A separate observer term is admissible only when it owns a disjoint apparatus field domain.

This gives an exact de-duplication rule at quadratic order. For a Hessian block

$$K = \begin{pmatrix} K_x & B \\ B^\dagger & K_y \end{pmatrix}, \quad K_{\text{eff}} = K_x - B K_y^{-1} B^\dagger, \quad (15)$$

one may retain the full block or eliminate y and retain its Schur reduction, but not both. Propagators and Green response operators are inverses of second variations of Eq. (14); they are not additional action or stress summands. Likewise, for any allowed finite local covariant counterterm $C_{\text{loc}}[e]$, the relocation

$$S_{\text{loc}}^{\text{ren}} \mapsto S_{\text{loc}}^{\text{ren}} + C_{\text{loc}}, \quad W_O^{\text{ren}} \mapsto W_O^{\text{ren}} - C_{\text{loc}} \quad (16)$$

leaves Γ_{MVP} and its full variations unchanged. Adding the same counterterm to both terms would change the equation and is double counting. Equations (12)–(16) prove a once-counted effective and quadratic ledger once a microscopic ownership partition and renormalization prescription are supplied. They do not construct the full nonlinear microscopic Tau action or its ultraviolet completion.

For an occupied observer carrier O , define the typed shared descriptor

$$\Xi_O = \text{Desc}_O(M_*, K_M, e, A, \psi, \vartheta, \{F_{O,a}\}). \quad (17)$$

More formally, on the adopted leaf

$$\text{Desc}_O : \mathcal{D}_{M_*} / \mathcal{N}_{\text{true}} \longrightarrow \mathcal{X}_O,$$

where $\mathcal{D}_{M_*}$ is the typed post-body data bundle and $\mathcal{X}_O$ is the shared observer descriptor space. Quotient safety requires $\text{Desc}_O(x) = \text{Desc}_O(x')$ whenever $x \sim_{\text{true}} x'$ under the retained common-null relation. The descent is the observer-specific realization of the ordering and compatibility law resident in the body. It is not a separate material layer, does not store a trajectory, and is not defined as a correction to a standard model. Standard physics and any future nonstandard component belong to the same full descent.

The terminal rows are generated by variations of the same packet:

$$\mathcal{E}_e = \frac{\delta \Gamma_{\text{MVP}}}{\delta e}, \quad \mathcal{E}_A = \frac{\delta \Gamma_{\text{MVP}}}{\delta A}, \quad (18)$$

$$p(a|O, M_*) = \left. \frac{\delta W_O}{\delta \eta_a} \right|_{\eta=0}, \quad \mathcal{E}_\vartheta = \frac{\delta \Gamma_{\text{MVP}}}{\delta \vartheta}. \quad (19)$$

Common ownership fixes cross-sector covariance and prevents duplicate matter or Green-response terms (Fig. 5).

10.4 Conditional gravity and optical leaf

On A1–A5 and A7, the low-curvature gravitational part of the completion has the effective form

$$\Gamma_g = \frac{1}{16\pi G_{\text{SHR}}} \int d^4x \sqrt{-g} (R - 2\Lambda_{\text{ren}}) + S_{\text{loc}}^{\text{ren}}[g] + W_O^{\text{ren}}[g, A, \Phi_{\text{ret}}, \eta, \vartheta; M_*], \quad (20)$$

where $S_{\text{loc}}^{\text{ren}}$ collects allowed local curvature counterterms and W_O^{ren} is the once-integrated quantum/observer functional. Coframe or metric stationarity gives

$$G_{\mu\nu} + \Lambda_{\text{ren}} g_{\mu\nu} + H_{\mu\nu}^{\text{loc,ren}} = 8\pi G_{\text{SHR}} \left(T_{\mu\nu}^{\text{body,ret}} + \langle T_{\mu\nu}^{q+O} \rangle_{\text{ren}} \right). \quad (21)$$

Here $T_{\mu\nu}^{\text{body,ret}}$ is the metric derivative of the retained body replacement term and excludes every degree of freedom integrated into W_O^{ren} . The second term is the metric derivative of that same effective functional and is not added as a separate microscopic action. Local covariance and diffeomorphism invariance give the corresponding Ward identity and conservation of the *complete* right-hand-side packet. The separate body and quantum pieces need not be conserved before their coupling and renormalization prescription are fixed. The structure of Eq. (21) is

J. One once-counted functional, several terminal variations

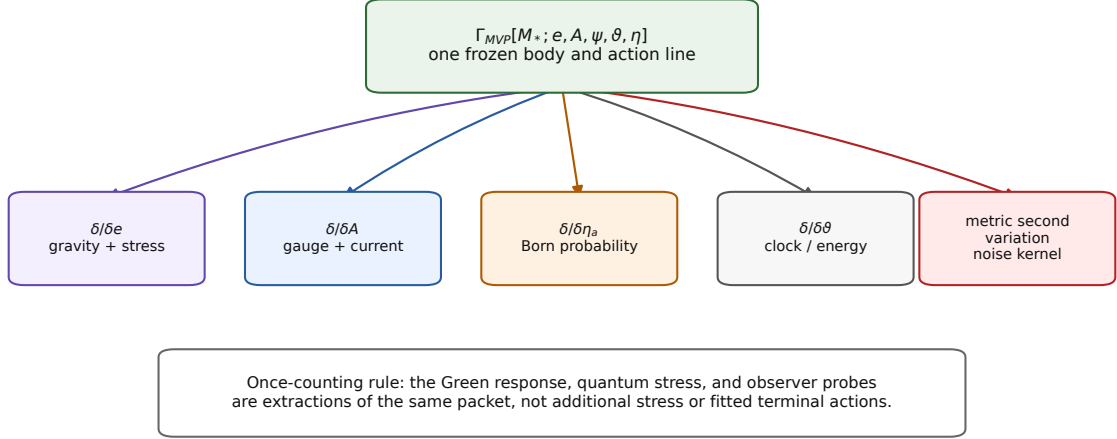


Figure 5: Terminal variations of one once-counted post-body functional. The diagram is a dependency map, not a claim that every displayed sector is already selected by the physical Tau parent.

the standard semiclassical-gravity structure, including finite local curvature ambiguities [41, 20, 21, 22]; the Tau claim is its conditional common-source placement, not novelty of the equation.

If quantum stress, stress fluctuations, and local higher-curvature terms are negligible, Eq. (21) reduces to the Einstein equation. Its weak-field, slow-motion leaf gives the Poisson equation and Newtonian motion. For a source-free Abelian radiation field, the same Lorentzian leaf gives the eikonal conditions

$$k^\mu k_\mu = 0, \quad k^\nu \nabla_\nu k^\mu = 0, \quad (22)$$

and the usual redshift and optical Jacobi equations. No independent lensing gain is introduced. These are conditional standard-limit recoveries; the completion does not predict numerical values of G_{SHR} or Λ_{ren} .

10.5 Conditional quantum and Born leaf

At fixed descended metric, let ρ_O be a positive normalized state and $\{F_{O,a}\}$ a positive complete effect family,

$$\rho_O \geq 0, \quad \text{Tr } \rho_O = 1, \quad F_{O,a} \geq 0, \quad \sum_a F_{O,a} = I. \quad (23)$$

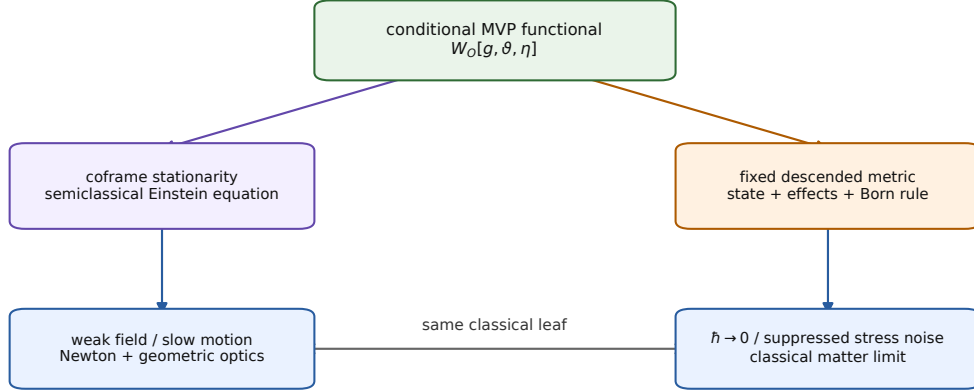
Under A6, complete additive noncontextual valuation yields the trace form

$$\boxed{p(a|O, M_*) = \text{Tr}(\rho_O F_{O,a}).} \quad (24)$$

This is a Gleason-type representation result [15, 8]. Its precise domain assumptions matter: ordinary projective Gleason theory requires Hilbert dimension at least three, while generalized-effect versions use stronger measurement richness to cover the two-dimensional case. Tau Core does not infer Eq. (24) from information loss alone; positivity, completeness, and valuation structure are indispensable assumptions of the frozen completion.

At fixed background and in a regular positive-frequency sector, the same functional gives the Schrödinger description and locally covariant quantum field theory. A Hadamard prescription supplies $\langle T_{\mu\nu}^{q+O} \rangle_{\text{ren}}$ in Eq. (21). Microcausality together with local completely positive

K. Compatible standard-limit diagram



Compatibility holds under the frozen regularity and limit-interchange assumptions; uniqueness does not follow.

Figure 6: Compatible standard leaves of the frozen completion. Correct limiting behavior is necessary but not sufficient for physical selection or novelty.

trace-preserving operations gives operational no-signalling; entanglement changes joint statistics without creating a controllable superluminal signal. These are recovery conditions shared with standard algebraic QFT in curved spacetime [7, 20, 21], not novel predictions of the Tau architecture.

10.6 Compatibility of the standard limits

The gravity and quantum leaves are not two separately fitted models. They are limits of Eq. (14). Holding the descended metric fixed gives the quantum/Born leaf. Varying the coframe while retaining the renormalized quantum expectation gives Eq. (21). Suppressing quantum stress fluctuations and then taking the classical matter limit reaches the same classical geometry as taking the classical limit first and varying the corresponding classical matter action.

Proposition 2 (Conditional compatibility of the standard leaves). *Under A1–A7, a common renormalization convention, and the stated regularity and limit–variation interchange assumptions, the fixed-background quantum limit and the small-fluctuation Einstein limit are compatible at the displayed semiclassical order. In the joint limit $\hbar \rightarrow 0$, suppressed stress noise, and negligible local higher-curvature terms, both routes reduce to classical matter on the same Einstein geometry.*

Conditional argument. Both routes are first variations of the same once-counted functional. The metric derivative of the renormalized matter term is the quantum stress in Eq. (21); replacing the state by its classical limit turns that derivative into classical matter stress. Because the body and renormalization convention are held fixed, no independent coefficient is introduced when the order of the two limits is exchanged. The stated regularity assumptions justify interchange of the limit and first variation. $\square$

This is a completion-internal consistency result, not a proof of nonperturbative quantum gravity, metric quantization, or uniqueness (Fig. 6).

10.7 QOR: a frozen observer-resolution discriminator

The MVP does not derive Lorentzian gravity or quantum state/effect structure from the weak Tau spine; it tests whether these adopted structures can be jointly organized as terminals of one frozen body-first completion. To do more than rewrite known equations, A8a-OCC together with A8b provides two source-frozen stable record prototypes $r_{O,0}$ and $r_{O,1}$ in the occupied observer carrier $\mathcal{C}_O$. Let $\iota_O : \mathcal{C}_O/N_O \rightarrow T_{M_*}\mathcal{M}/\mathcal{N}_{\text{true}}$ be its source-declared linearized embedding into the retained body tangent space. The pullback of the positive body Hessian is

$$G_O = \iota_O^* K_M \iota_O, \quad \mathcal{A}_X^{PC} = \langle \Delta r_O, G_O \Delta r_O \rangle, \quad \Delta r_O = r_{O,1} - r_{O,0}. \quad (25)$$

The expression is coordinate independent and quotient-safe when ι_O descends to the displayed quotient. It has units of action because K_M is the body action Hessian. A8a-OCC and the support theorem below supply the common amplitude carrier, while A8b supplies the finite global completion of this local metric. Here ‘‘Gram amplitude’’ means an Uhlmann amplitude W satisfying $WW^\dagger = \rho$. Let normalized amplitudes $W_{O,i}$ satisfy $W_{O,i}W_{O,i}^\dagger = \rho_{O,i}$ and $\text{Tr } \rho_{O,i} = 1$. The same inherited action metric is used on their right-unitary quotient, with no independent coefficient:

$$\frac{\mathcal{A}_X^{PC}}{\mathcal{A}_*} = \min_U \|W_{O,1} - W_{O,0}U\|_{\text{HS}}^2, \quad U^\dagger U = UU^\dagger = I. \quad (26)$$

The amplitudes are taken in a common sufficiently large purification space, so the finite-dimensional optimization can be written over right unitaries. If minimal purification spaces have unequal support, the equivalent partial-isometry formulation is understood; an infinite-dimensional extension would require separate domain and trace-class control.

Lemma 1 (Local metric of the common-Gram minimum lift). *Assume constant rank in a neighborhood of q_0 and choose a smooth local horizontal gauge. Let $W(q)$ be a twice-differentiable normalized amplitude curve through $W(q_0)$, and let $\dot{W}_h$ be its horizontal tangent after minimizing the vertical right-unitary direction. Then*

$$\min_U \|W(q_0 + \delta q) - W(q_0)U\|_{\text{HS}}^2 = \delta q^2 \|\dot{W}_h\|_{\text{HS}}^2 + O(|\delta q|^3). \quad (27)$$

Under A8b, the pullback of this quotient metric is $G_O/\mathcal{A}_*$.

Proof. Write a nearby gauge representative as $W(q_0 + \delta q)U(\delta q)^\dagger = W(q_0) + \delta q(\dot{W} - W(q_0)X) + O(\delta q^2)$, with $X^\dagger = -X$. Minimizing the quadratic coefficient over X removes the vertical tangent and leaves $\dot{W}_h$. Squaring the Hilbert–Schmidt norm gives Eq. (27). A8b identifies this horizontal quotient norm, with the common action unit restored, with the body-Hessian pullback in Eq. (25). The last identification is therefore explicit completion structure, not a consequence of Hessian positivity alone. $\square$

A8b is the strongest physical identification in the QOR completion. The lemma proves the local metric of the Uhlmann quotient; it does not prove that the Tau body Hessian owns that metric. Body stationarity, Hessian positivity, endpoint blindness, and readout emergence do not by themselves imply A8b.

Let $\mathcal{A}_* > 0$ be the common elementary action scale and define

$$\epsilon_O^2 = \frac{\mathcal{A}_X^{PC}}{\mathcal{A}_*} > 0. \quad (28)$$

Both numerator and denominator have the same units, so ϵ_O is dimensionless. The completion identifies the common action unit with $\hbar$ on the quantum leaf. It then declares a dimensionless quantum-state angle

$$DQ(\rho_0, \rho_1) = \begin{cases} \arccos |\langle \psi_0 | \psi_1 \rangle|, & \rho_i = |\psi_i\rangle\langle \psi_i|, \\ \arccos \sqrt{F(\rho_0, \rho_1)}, & \text{mixed states,} \end{cases} \quad (29)$$

where

$$F(\rho, \sigma) = \left[\text{Tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right]^2$$

is the Uhlmann fidelity. Uhlmann’s minimum-lift identity [40] applied to Eq. (26) gives

$$\epsilon_O^2 = 2 \left(1 - \sqrt{F(\rho_{O,0}, \rho_{O,1})} \right) = 4 \sin^2 \frac{D_Q(\rho_{O,0}, \rho_{O,1})}{2}. \quad (30)$$

Consequently

$$D_Q(\rho_{O,0}, \rho_{O,1}) = 2 \arcsin \frac{\epsilon_O}{2} \geq \epsilon_O, \quad 0 < \epsilon_O \leq \sqrt{2}. \quad (31)$$

The final inequality uses $2 \sin(x/2) \leq x$ on $0 \leq x \leq \pi/2$. It is therefore a theorem of the common-Gram completion, not a separate distance assumption. Positivity of K_M alone would still be insufficient: the decisive input is the global quotient minimum-lift identity, which already includes all purification shortcuts.

For self-adjoint unitary transport, the Fubini–Study/Bures path obeys

$$D_Q(\rho(0), \rho(\delta t)) \leq \frac{1}{\hbar} \int_0^{\delta t} \Delta H(t) dt, \quad \Delta H(t)^2 = \text{Tr}(\rho H^2) - \text{Tr}(\rho H)^2, \quad (32)$$

with mixed-state extensions expressed through the corresponding quantum Fisher information [25, 3, 36]. Equation (32) is restricted here to the declared self-adjoint unitary protocol. Open or noisy evolution requires its own quantum-Fisher-information path-speed bound and cannot inherit the same ΔH formula without an additional proof. For constant ΔH , combining Eqs. (31) and (32) gives the exact common-Gram bound

$$\boxed{\delta t_{\min, O} \geq \frac{2\hbar}{\Delta H} \arcsin \frac{\epsilon_O}{2} \geq \frac{\hbar \epsilon_O}{\Delta H}.} \quad (33)$$

The inverse-energy-spread form and the Uhlmann chord-to-angle identity are standard, not new Tau laws. The distinct Tau statement is narrower: the source-frozen morphological Gram chord is physically identified with the observer carrier’s Uhlmann-amplitude distance. At the same ΔH , two independently reconstructed observer carriers with different source-frozen ϵ_O have different forbidden single-record regions, provided the common-Gram carrier is physically occupied. The numerical prefactor is fixed by the angle convention in Eqs. (29)–(32). An ordering of *observed* onset thresholds follows only if the bound is saturated, or if independently controlled efficiencies establish the same slack above the bound. No detector gain may be fitted to enforce either conclusion.

For an empirical test, “independent reconstruction” therefore has a precise meaning: the carrier class, the two record prototypes, the embedding ι_O , and the pullback metric G_O must be fixed from source-side morphology and calibration data that exclude the timing endpoint. If these objects cannot be operationalized independently, Eq. (33) remains a conditional mathematical discriminator rather than a test-ready numerical prediction.

Here a *single stable record* means that one predeclared protocol run places a pointer variable in one of two disjoint decision regions with single-shot classification error at most η_{rec} , and that the assignment persists for a predeclared hold time τ_{hold} . No averaging across repeated preparations, tomographic reconstruction, or multi-shot parameter estimation is part of that record event. Those methods may calibrate the carrier or estimate sub-threshold changes, but they test a different operational question.

Finite carrier prototype and uncertainty ledger. A two-level carrier makes the full mathematical chain explicit without claiming physical occupation. The full qubit state carrier

is the density- operator space $\mathcal{D}(\mathbb{C}^2)$; $\mathbb{CP}^1$ is only its pure- state projective submanifold. Restrict the prototype records to $\mathcal{C}_O^{\text{pure}} = \mathbb{CP}^1$, choose $\rho_{O,0} = |0\rangle\langle 0|$, and set

$$|\psi_{O,1}\rangle = \cos \theta_O |0\rangle + \sin \theta_O |1\rangle, \quad \rho_{O,1} = |\psi_{O,1}\rangle\langle \psi_{O,1}|.$$

Then $D_Q(\rho_{O,0}, \rho_{O,1}) = |\theta_O|$ for $0 \leq |\theta_O| \leq \pi/2$. The common-Gram identity gives the exact finite separation

$$\epsilon_O = 2 \sin \frac{|\theta_O|}{2}, \quad |\theta_O| = 2 \arcsin \frac{\epsilon_O}{2} \geq \epsilon_O. \quad (34)$$

Thus the two-level example realizes the global theorem rather than assuming a second distance calibration. If a source-declared carrier coordinate q has tangent image $v_O = D\iota_O(\partial_q)$, set $g_O = \langle v_O, K_M v_O \rangle$. In a normal neighborhood,

$$\epsilon_O = |\Delta q| \sqrt{\frac{g_O}{\mathcal{A}_*}} + O(|\Delta q|^2). \quad (35)$$

The local Hessian estimate is not substituted for the exact finite minimum-lift distance outside that neighborhood.

To make “phase-calibrated quasistatic response” reproducible, choose before the timing experiment a closed calibration protocol $\mathcal{P}_{\text{cal}}$ and define its unwrapped action phase by $\phi_{\text{cal}}(q) = \mathcal{A}_{\text{cal}}(q)/\hbar$. Near the source-declared reference q_0 , the operational model is

$$\mathcal{A}_{\text{cal}}(q) = \mathcal{A}_0 + f_0(q - q_0) + \frac{1}{2}g_O(q - q_0)^2 + O((q - q_0)^3). \quad (36)$$

For a symmetric predeclared step h_q , an endpoint-blind curvature estimator is

$$\frac{\widehat{g}_O}{\hbar} = \frac{\phi_{\text{cal}}(q_0 + h_q) - 2\phi_{\text{cal}}(q_0) + \phi_{\text{cal}}(q_0 - h_q)}{h_q^2} + O(h_q^2). \quad (37)$$

Equivalently, if $f_q = \partial_q \mathcal{A}_{\text{cal}}$ is measured, then $g_O = \partial_q f_q|_{q_0}$. This equality follows rather than being an independent calibration postulate when the protocol is the restriction of the same body action to a source-predeclared curve $M(q)$, with $M(q_0) = M_*$ and $v_O = M'(q_0)$:

$$\left. \frac{d^2}{dq^2} \mathcal{A}_{\text{body}}(M(q)) \right|_{q_0} = K_M(v_O, v_O) + D_M \mathcal{A}_{\text{body}}(M_*)[M''(q_0)] = g_O. \quad (38)$$

The final equality uses body stationarity. The neighboring calibration configurations are source-prepared before the timing experiment; this is not terminal backreaction on the already frozen body. Failure of Eq. (38) rejects the proposed common-source carrier realization rather than authorizing a refit.

An endpoint-free calibration must therefore fix the carrier, q , q_0 , h_q , phase-unwrapping rule, and $\mathcal{P}_{\text{cal}}$ from non-timing device morphology; estimate $g_O/\mathcal{A}_*$ with Eq. (37); determine D_Q by state tomography; and determine ΔH by independent energy statistics. The common-Gram identity and Eq. (38) can then be tested before the timing endpoint is opened, avoiding conditional circularity. For a local reconstruction with independent Δq , g_O , and $\mathcal{A}_*$ estimates, first-order uncertainty propagation gives

$$\left(\frac{\sigma_\epsilon}{\epsilon_O} \right)^2 = \left(\frac{\sigma_{\Delta q}}{\Delta q} \right)^2 + \frac{1}{4} \left(\frac{\sigma_g}{g_O} \right)^2 + \frac{1}{4} \left(\frac{\sigma_{\mathcal{A}_*}}{\mathcal{A}_*} \right)^2, \quad (39)$$

or $\sigma_\epsilon^2 = J \Sigma J^T$ with the full calibration covariance. Metrologically $\hbar$ is exact in SI, while its identification with the parent action unit remains a physical assumption of the completion. Two carriers are comparable only under the same Hamiltonian generator and matched ΔH protocol; otherwise the predeclared comparison variable is $\Delta H \delta t/\hbar$, not raw time.

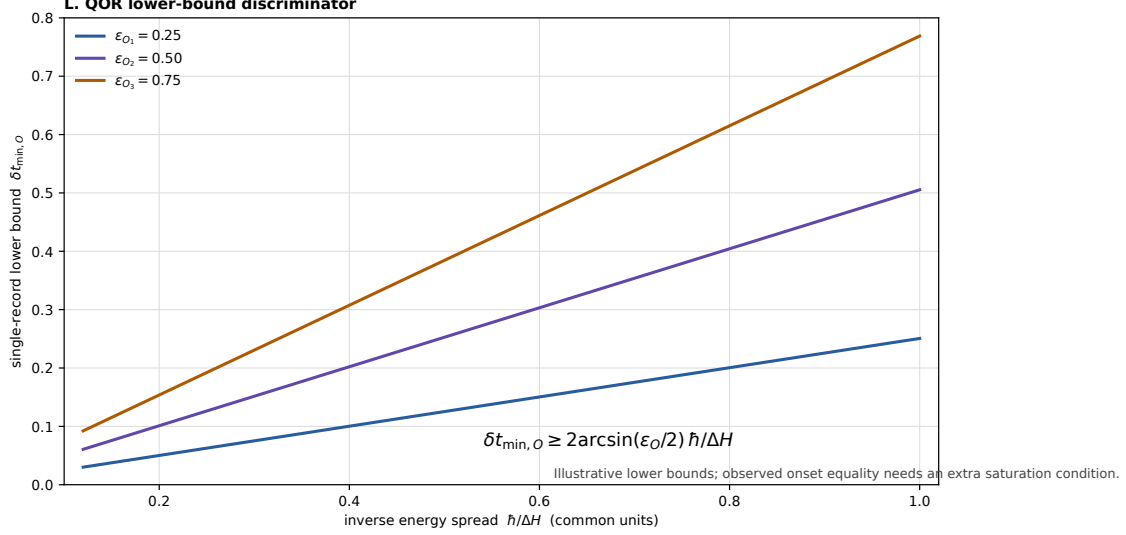


Figure 7: Illustration of Eq. (33). The slopes are not fitted or predicted numerical values. They show the ordering and ratio of *lower-bound* slopes after ϵ_O has been reconstructed independently. Equality of these lines with observed onset thresholds requires an additional saturation or controlled-efficiency condition.

This prototype closes the type, normalization, distance bridge, and uncertainty bookkeeping. It also reduces pullback-stiffness reconstruction to the same-body action identity (38). It does not prove that the physical Tau source occupies this common-Gram carrier or supplies a laboratory realization of $\mathcal{C}_O$ and ι_O .

Equation (33) is not a universal time atom and permits continuous parameters. Aggregation may resolve sub-record changes. A test must therefore predeclare the carrier, record criterion, ΔH , aggregation rule, and estimator. Figure 7 shows bounds, not predicted onsets.

Minimal falsification protocol. The test is staged so that a failed carrier is rejected before timing data are used. On one predeclared carrier coordinate and record pair, define

$$\hat{x}_A = \frac{\widehat{\mathcal{A}_X^{PC}}}{\mathcal{A}_*}, \quad \hat{x}_Q = 2 \left(1 - \sqrt{\widehat{F}(\rho_{O,0}, \rho_{O,1})} \right). \quad (40)$$

Stage 1 reconstructs $\hat{x}_A$ from the closed same-action curvature/work protocol and checks Eq. (38), without tomography or endpoint timing. Stage 2 reconstructs $\hat{x}_Q$ by independent state tomography and tests the Tau-specific identity

$$\boxed{\hat{x}_A \stackrel{?}{=} \hat{x}_Q}. \quad (41)$$

For independent estimates, a minimal discrepancy statistic is

$$Z_A = \frac{\hat{x}_Q - \hat{x}_A}{\sqrt{\sigma_{x_Q}^2 + \sigma_{x_A}^2}}, \quad (42)$$

with the full declared covariance used when systematic errors are correlated. Defining either estimator from the other makes the test circular. Only a carrier passing both pre-timing tests proceeds to Stage 3, the QOR timing test. Thus failure of the common-Gram realization cannot be hidden by a timing fit.

Choose at least two observer carriers with independently reconstructed $\epsilon_{O_1} \neq \epsilon_{O_2}$, hold the Hamiltonian-spread and record protocol fixed, and estimate the single-record onset without using the timing endpoint in carrier selection. A reproducible stable record at $\delta t < 2\hbar \arcsin(\epsilon_O/2)/\Delta H$, after demonstrating that the one-record, common-Gram minimum-lift, source-freeze, and self-adjoint unitary-protocol conditions hold, rejects the completion. A cross-carrier onset ordering is an additional test only after saturation or equal-slack conditions have been independently justified. Failure of this completion does not by itself reject every possible Tau-style readout architecture.

10.8 Finite source criterion and carrier decision

The two parts of A8 can be reduced to finite source conditions rather than left as an undifferentiated bridge assumption. Let $K_* > 0$ be the body Hessian after the true-null quotient and let

$$R_{\text{rec}}^{\text{dedup}} : N_*^{\text{phys}} \longrightarrow P_{\text{rec}} \mathcal{H}_{\text{rec}}$$

be the de-duplicated primitive recovery incidence. Define

$$G_*^{\text{rec}} = R_{\text{rec}}^{\text{dedup}} K_*^{-1} (R_{\text{rec}}^{\text{dedup}})^{\dagger}, \quad \rho_*^{\text{rec}} = \frac{G_*^{\text{rec}}}{\text{Tr } G_*^{\text{rec}}}. \quad (43)$$

Proposition 3 (Support-restricted common-amplitude theorem). *Let $K_* > 0$ on a finite retained source tangent space and let $R_{\text{rec}}^{\text{dedup}} \neq 0$. Set*

$$P_* = \text{supp } G_*^{\text{rec}}. \quad (44)$$

Then ρ_^{rec} is faithful on $P_* \mathcal{H}_{\text{rec}}$, whether or not the recovery incidence is onto a larger declared ambient space. If one source-owned CP record instrument $\{\mathcal{I}_i\}$ acts on this occupied support and*

$$\rho_i = \frac{\mathcal{I}_i(\rho_*^{\text{rec}})}{\text{Tr } \mathcal{I}_i(\rho_*^{\text{rec}})} \neq 0, \quad (45)$$

then all ρ_i admit Uhlmann amplitudes in the same Hilbert–Schmidt carrier. A smooth local amplitude bundle exists on every constant-rank stratum.

Proof. For every y ,

$$\langle y, G_*^{\text{rec}} y \rangle = \left\| K_*^{-1/2} (R_{\text{rec}}^{\text{dedup}})^{\dagger} y \right\|^2,$$

so $G_*^{\text{rec}} \succeq 0$ and $\ker G_*^{\text{rec}} = \ker (R_{\text{rec}}^{\text{dedup}})^{\dagger}$. Restriction to $P_* \mathcal{H}_{\text{rec}}$ is therefore strictly positive. Functional calculus gives the canonical amplitude $W_* = \sqrt{\rho_*^{\text{rec}}}$, and polar decomposition gives every minimal presentation as $W_* U$ up to the usual partial-isometry extension. A CP map has positive output in the same support algebra, so every nonzero conditional state in Eq. (45) has a square-root amplitude in the same Hilbert–Schmidt carrier. Smooth functional calculus and a smooth horizontal gauge give the local bundle at constant rank. $\square$

Thus common-amplitude existence is kinematic once the occupied support and record instrument are supplied. Ambient surjectivity and uniqueness among all stationary parent branches are not needed for this theorem; they would be stronger nature-selection claims. What remains open in A8a-OCC is precisely whether the physical Tau parent occupies this support/instrument packet.

Record-domain and occurrence boundary. No additional apparatus ontology is needed to type one finite record once a positive effect is supplied. For any $0 \preceq E_O \preceq I$, introduce a two-state pointer initialized in $|0\rangle_P$ and set

$$\alpha_O = \arcsin \sqrt{E_O}, \quad U_{E_O} = \exp(-i\alpha_O \otimes \sigma_y). \quad (46)$$

Projection onto the pointer basis gives

$$K_1 = {}_P\langle 1|U_{E_O}|0\rangle_P = \sqrt{E_O}, \quad K_0 = {}_P\langle 0|U_{E_O}|0\rangle_P = \sqrt{I - E_O}, \quad (47)$$

up to an irrelevant branch phase. Thus one supplied occurrence produces the canonical square-root CP instrument without a fitted probability gain. Likewise, for any finite orthogonal pointer algebra and supplied record incidences T_j , the positive matching functional

$$\mathcal{A}_{\text{rec}}(r_1, \dots, r_m \mid p) = \frac{1}{2} \sum_{j=1}^m \kappa_j \|r_j - T_j p\|^2, \quad \kappa_j > 0, \quad (48)$$

has Hessian $\bigoplus_j \kappa_j I > 0$ and the unique structural minimum $r_j = T_j p$. The construction is natural in the finite pointer dimension; it adds neither a Born weight nor a new stiffness law.

Equations (46)–(48) prove availability and stability *conditional on an occurrence*. They do not select whether an observer–record edge is physically occupied. The same static carrier, effect, clock, and conserved current admit a zero-occurrence model and a one-occurrence model. Effect normalization fixes probability conditional on an interrogation, while carrier flux counts occupation rather than interrogations. A finite observer-time persistence interval is also stronger than positivity of the record Hessian: it requires a descended retention generator or direct operational verification. Hence A8a-OCC is reduced to a physical incidence statement, not a missing record-domain type.

A8b has two logically distinct defects. Let $\mathcal{C} : (N_*^{\text{phys}}, K_*) \rightarrow (\mathcal{E}_*, h_*)$ be the proposed body-to-amplitude differential and let P_O be the h_* -orthogonal observer-access projector. Define

$$E_{\text{src}} = \mathcal{C}^\dagger h_* \mathcal{C} - K_*, \quad \Delta_O = \mathcal{C}^\dagger h_* (I - P_O) \mathcal{C} \succeq 0. \quad (49)$$

The actually descended metric is therefore

$$G_O = (P_O \mathcal{C})^\dagger h_* (P_O \mathcal{C}) = K_* + E_{\text{src}} - \Delta_O. \quad (50)$$

The no-loss statement

$$\Delta_O = 0 \iff \text{im } \mathcal{C} \subseteq \text{im } P_O \quad (51)$$

does not prove $E_{\text{src}} = 0$. A8b follows without a fitted scale only if one and the same parent action supplies the normalized connector identities

$$\boxed{\mathcal{C}^\dagger h_* \mathcal{C} = K_*, \quad P_O \mathcal{C} = \mathcal{C}.} \quad (52)$$

The first identity is source isometry; the second is complete access only on the selected record tangent, not on the whole parent body. Under both identities, quotienting the right-unitary vertical directions gives the Hilbert–Schmidt/Uhlmann metric with coefficient exactly $K_*/\mathcal{A}_*$. Multiplying $\mathcal{C}$ by a positive constant would change the first identity and is a different physical source incidence, not a coordinate convention.

Neither common support nor injective conformal descent selects this identity. Indeed, $\mathcal{C}_c = \sqrt{c} U$, with U an isometry and $c > 0$, has the same rank, complex structure, conformal cone, and normalized clock-line transport for every c , while its pullback metric changes by c . Consequently A8a-OCC and Hessian positivity do not imply A8b.

The consistency of A8b can nevertheless be exhibited without a fitted coefficient. Write

$$x_O = \frac{\mathcal{A}_X^{PC}}{\mathcal{A}_*}. \quad (53)$$

For every $0 \leq x_O \leq 2$, define

$$r(x_O) = \sqrt{x_O - \frac{x_O^2}{4}}, \quad \rho_{\pm}(x_O) = \frac{1}{2} (I \pm r(x_O) \sigma_z). \quad (54)$$

These are normalized positive qubit states and obey

$$2 \left(1 - \sqrt{F(\rho_+, \rho_-)} \right) = x_O. \quad (55)$$

Thus A8b has a normalized mixed-state realization exactly on $0 \leq x_O \leq 2$; distinct records require $0 < x_O \leq 2$. It is also a valid CP observer restriction: the depolarizing map $\mathcal{E}_r(\rho) = r\rho + (1-r)I/2$ sends an orthogonal qubit pair to Eq. (54) and admits a standard Stinespring isometry V_r .

This proves non-emptiness, not nature-level selection. The current body/Gibbs and CP-descent laws admit every $r \in [0, 1]$ and contain no term that chooses its value. The remaining physical statement is therefore the finite source identity

$$\boxed{V_{\text{phys}} = V_{r(x_O)}, \quad r(x_O)^2 = x_O - \frac{x_O^2}{4}, \quad 0 < x_O \leq 2.} \quad (56)$$

It must be a second restriction of the same post-body functional that owns x_O . Choosing r from the timing endpoint or measured record overlap would be terminal fitting. Hence A8b is a mathematically closed conditional completion, while its physical source selection remains open.

Proposition 4 (Record occurrence and finite A8b are independent). *Relative to assumptions A1–A9 without an additional law coupling occurrence degree to the amplitude connector, physical record occurrence and the finite A8b identity are logically independent.*

Proof. Fix $x_O \in (0, 2]$ and define two propositions

$$\mathbf{O} : n_{\text{rec}, O} \geq 1, \quad \mathbf{B} : 2(1 - \sqrt{F(\rho_+, \rho_-)}) = x_O, \quad (57)$$

where $n_{\text{rec}, O}$ counts source-owned record incidences in the declared observer support. The depolarizing Stinespring family used above is valid for every $r \in [0, 1]$, and its symmetric output pair has squared Uhlmann chord

$$y(r) = 2 \left(1 - \sqrt{1 - r^2} \right). \quad (58)$$

The value $r_*(x_O)$ in Eq. (54) satisfies $y(r_*) = x_O$. Choose any $r_0 \in (0, 1]$ with $r_0 \neq r_*$ and $y(r_0) \neq x_O$. The models $(n_{\text{rec}, O}, r) = (1, r_*)$ and $(1, r_0)$ have the same body ratio, finite pointer type, occurrence degree, and positive record-matching Hessian, but only the first satisfies **B**. Therefore $\mathbf{O} \not\Rightarrow \mathbf{B}$.

Conversely, hold the carrier geometry fixed at $r = r_*$ and compare $(n_{\text{rec}, O}, r) = (1, r_*)$ with $(0, r_*)$. The finite metric identity holds in both admissible carrier completions, while only the first contains a physical record event. Thus $\mathbf{B} \not\Rightarrow \mathbf{O}$. Repeating the zero/one occurrence choice at r_0 realizes all four truth assignments of $(\mathbf{O}, \mathbf{B})$, proving independence. The zero-occurrence A8b case is a structural law on an available carrier, not a measured record. $\square$

The proposition is relative to the current source axioms, not a prohibition on deeper unification. A richer post-body source law may select occurrence and the connector jointly. In the present completion, however, a realized QOR trial requires the conjunction

$$\boxed{n_{\text{rec},O} \geq 1 \quad \wedge \quad x_A = x_Q \quad \wedge \quad H_{\text{hold}}.} \quad (59)$$

Here H_{hold} denotes the independently predeclared single-record retention condition. Occupation supplies an event, A8b supplies its finite morphology/Bures calibration, and the hold condition makes the event operationally readable; none substitutes for another.

10.9 A finite source theorem for A8b

Let

$$\mathfrak{A}_{\text{rec}} \subseteq \text{End}(\mathcal{H}_{\text{rec}}) \quad (60)$$

be the finite unital star algebra generated by the source-owned primitive recovery relations of the stabilized body, and let $Z_\alpha \in Z(\mathfrak{A}_{\text{rec}}'')$ be mutually orthogonal central recovery projectors with $\sum_\alpha Z_\alpha = P_{\text{rec}}$.

Central recovery support and faithful occupation do not make these projectors reducing for an independently supplied Dirac operator. On $\mathbb{C}^2$, keep the diagonal recovery algebra, its two rank-one central projectors, and the faithful state $I/2$ fixed. The pair

$$D_0 = \begin{pmatrix} d_1 & 0 \\ 0 & d_2 \end{pmatrix}, \quad D_a = \begin{pmatrix} d_1 & a \\ a & d_2 \end{pmatrix}, \quad a \neq 0, \quad (61)$$

has identical recovery-algebra and occupation data, whereas only D_0 commutes with the central projectors. Recovery cyclicity and central support therefore do not imply a Dirac direct sum.

Proposition 5 (Recovery–Dirac affiliation implies finite A8b). *Assume A8a-OCC and the common-action recovery construction. If a self-adjoint global Dirac operator D is affiliated with $\mathfrak{A}_{\text{rec}}''$, and the parent action uses one fixed spectral profile, spectral scale, and trace normalization on the occupied recovery support, then the central recovery projectors reduce D , the spectral action is additive over the recovery blocks, and finite A8b follows without an additional gain.*

Proof. Affiliation means that every spectral projector of D belongs to $\mathfrak{A}_{\text{rec}}''$. Every central Z_α therefore strongly commutes with D , and

$$P_{\text{rec}} D P_{\text{rec}} = \bigoplus_\alpha D_\alpha, \quad D_\alpha = Z_\alpha D Z_\alpha. \quad (62)$$

For one fixed profile F and scale Λ , functional calculus and trace additivity give

$$\text{Tr } F(D^2/\Lambda^2) = \sum_\alpha \text{Tr } F(D_\alpha^2/\Lambda^2). \quad (63)$$

This is the exclusive orthogonal direct-sum composition required by the same-action recovery theorem. Squared Hilbert–Schmidt/Uhlmann chords then add in the common action unit, giving finite A8b. $\square$

This is a sufficient source theorem, not a proof that the physical parent occupies its premises. In general the body Hessian and D act on different spaces, so a bare commutator $[K_M, D]$ is not the source condition. The body Hessian is a pullback of the spectral Hessian through the body-to-Dirac derivative, including its second-derivative chain-rule term. A spectral trace can therefore retain cross-block body terms unless the incidence is source-compatible. Affiliation resolves this issue for central recovery blocks because the operator and the recovery decomposition are generated by the same algebra.

The remaining question is one physical ownership statement: does the enriched base–seed parent construct its global Dirac operator as an operator affiliated with $\mathfrak{A}_{\text{rec}}''$? Equation (61) proves that the current source axioms do not answer it.

At quadratic response order the once-counted microscopic source ledger is explicit:

$$\Gamma_{\text{src}}^{(2)}[x; J] = \Gamma_* + \frac{1}{2} \langle x, K_* x \rangle - \text{Re} \langle J, R_{\text{rec}}^{\text{dedup}} x \rangle. \quad (64)$$

Eliminating x gives G_*^{rec} as the connected source two-point response. In a finite inner-product realization, overlap removal can be written

$$R_{\text{rec}}^{\text{dedup}} = \begin{pmatrix} R_{P1} \\ R_Q(I - R_{P1}^+ R_{P1}) \end{pmatrix}, \quad (65)$$

so a quantum row already represented by the common gravity–time–radiation–matter packet is not counted again. Equations (64)–(65) close a finite second-variation ledger, not the full interacting continuum Γ_{MVP} . The latter still requires a microscopic nonlinear base–seed action, interacting matter and gauge spectrum, and a controlled renormalized continuum limit.

Dynamic and quantized metrics. For a classical dynamic metric, locally covariant QFT assigns an observable algebra, positive normalized state, and effects to each admissible geometry; covariant unital star-morphisms preserve the state/effect probability form [7]. The state, effect, and propagator may all depend on the metric, and no preferred particle splitting is implied, but the Born form itself survives. A classical probability mixture over geometries also preserves positivity and normalization, but the branchwise Born laws do not determine the mixing measure.

If the metric is quantized, the corresponding statement requires a constrained joint geometry–matter algebra or physical Hilbert space with a positive normalized joint state and joint effects. Matter-only probabilities are then restrictions or conditional probabilities. This is compatible with Born structure, but Tau Core does not yet provide the metric measure, gauge/BRST–BV quotient, physical inner product, coherent inter-geometry phases, or positive unitary descent needed to construct it. This gap cannot be filled by applying a fixed-fiber Gleason/Busch theorem branch by branch: two coherent joint states may have the same block-diagonal geometry and matter data but different relative phases, which a joint off-diagonal effect distinguishes. The present Born recovery therefore covers fixed or classical-dynamic descended geometry. Quantized geometry remains a conditional joint-algebra completion, not a result of this paper.

A non-genericity control for candidate carriers. Quantum geometry is measurable in superconducting circuits and spin platforms [37, 42], so these systems are plausible worked-model platforms. They do not automatically satisfy A8b. For a nondegenerate ground state of $H(q)$, standard perturbation theory gives

$$g_{qq}^{\text{FS}} = \sum_{n \neq 0} \frac{|\langle n | \partial_q H | 0 \rangle|^2}{(E_n - E_0)^2}, \quad (66)$$

$$\partial_q^2 E_0 = \langle 0 | \partial_q^2 H | 0 \rangle + 2 \sum_{n \neq 0} \frac{|\langle n | \partial_q H | 0 \rangle|^2}{E_0 - E_n}. \quad (67)$$

The denominators and contact term differ; fidelity susceptibility and energy curvature are related probes but are not generally identical [1]. For

$$H(q) = \frac{\Delta}{2} (\cos q \sigma_z + \sin q \sigma_x),$$

the ground-state Fubini–Study metric is $1/4$, whereas the ground energy is constant and its curvature vanishes. Thus an ordinary energy-Hessian calibration fails A8b even in this elementary carrier. A physical Tau carrier must instead demonstrate that an independently specified closed *action* protocol obeys Eqs. (52) and (56). No currently identified platform has passed the curvature, Gram/Bures, and QOR timing tests together.

Effect-size boundary. For a resonant qubit with $\Delta H = \hbar\Omega/2$, Eq. (33) becomes

$$\tau_{\text{QOR}} = \frac{4}{\Omega} \arcsin \frac{\epsilon_O}{2}, \quad \frac{\tau_{\text{QOR}}}{t_\pi} = \frac{4}{\pi} \arcsin \frac{\epsilon_O}{2}. \quad (68)$$

It ranges from zero to one π -pulse for $0 \leq \epsilon_O \leq \sqrt{2}$. Detectability requires at least $\sigma_t \ll \tau_{\text{QOR}} \ll T_2$, record persistence below the relevant relaxation scale, and the declared single-shot error ceiling. However, Tau Core currently predicts no physical source-side value of ϵ_O . For an ideal timing resolution σ_t , inversion of Eq. (68) gives only the instrumental sensitivity floor

$$x_{\min}^{(t)} = 4 \sin^2 \left(\frac{\Omega \sigma_t}{4} \right) \simeq \frac{\Omega^2 \sigma_t^2}{4}, \quad (69)$$

not a predicted Tau effect size. Moreover, after A8 has identified ϵ_O with the Uhlmann chord, Eq. (33) is the standard Bures-angle quantum speed limit, not an additional delay. The Tau-specific test is therefore the independently predicted equality between body-action curvature and quantum-state distance. Without that source-side prediction, the timing stage is a conditional consistency or falsification check, not a distinctive positive signal that can yet be compared numerically with decoherence.

10.10 Exact open step: physical selection

The internal chain ends at Eq. (33); the physical chain fails one step earlier. In present notation,

$$B_\tau^{\text{phys}} + s_U \not\Rightarrow_{\text{current derivation}} \Gamma_{\text{MVP}} + \text{state/effect completion} + \text{QOR carrier}. \quad (70)$$

Proposition 6 (Upstream non-uniqueness within the admitted class). *The current base–seed–body reduct does not uniquely select the frozen MVP completion.*

Countermodel sketch. Keep the same stabilized body, quotient metric, Lorentzian infrared leaf, and classical coframe response. On a three-dimensional internal Hilbert factor, choose two orthogonal density matrices with matched classical stress but different probabilities for a fixed separating effect. Both are positive, normalized, and obey the same trace-form valuation, while the upstream body data do not select between them. Independently, two positive observer completions may use different nonzero record separations without changing the same classical coframe leaf. The resulting completions disagree on Eq. (24) or ϵ_O while preserving the admitted upstream reduct and standard classical limit. Hence identical upstream data have inequivalent terminal completions. Unique physical selection is therefore not established, which is precisely the non-implication recorded in Eq. (70). $\square$

This negative result is central. It prevents the conditional completion from being advertised as a first-principles theory of Nature. The remaining binary question is whether a deeper source law physically occupies the frozen completion. That question may be answered by a source theorem or by the source-frozen test of Eq. (33), but not by adding another fitted terminal.

10.11 Finite source-signature selection theorem

The non-uniqueness result identifies what the current upstream reduct lacks, but it does not require an arbitrary list of additional terminal parameters. Suppose the enriched source supplies a finite-dimensional unital C^* -algebra $\mathcal{A}_s \subseteq \text{End}(\mathcal{H}_s)$, generated by its typed relation incidences after the true source-null quotient. On its occupied recovery ideal let

$$G_s = R_s^{\text{dedup}} K_*^{-1} (R_s^{\text{dedup}})^\dagger, \quad \rho_s = \frac{G_s}{\text{Tr } G_s}, \quad \omega_s(a) = \text{Tr}(\rho_s a). \quad (71)$$

If R_s^{dedup} is onto that ideal, then $G_s > 0$ and ω_s is faithful there.

Let $\mathbf{z}_s$ be the finite tuple of all source-owned distinguished coframe, gauge, matter, clock, effect, record, hold, A8, and ownership elements or CP maps. In a source-orthonormal basis $e_1, \dots, e_d$, define the structured signature

$$\Sigma_s^{\text{GNS}} = \left(c_{ab}^c, j_a^b, \omega_s(e_a), [z_\alpha]_a, [\mathcal{I}_\beta]_{ab} \right), \quad e_a e_b = c_{ab}^c e_c, \quad e_a^* = j_a^b e_b. \quad (72)$$

Simultaneous basis changes are quotiented as typed source isometries.

Proposition 7 (Finite source-signature uniqueness). *Two minimal cyclic finite completions carrying the same Σ_s^{GNS} are unitarily equivalent by the unique unitary that maps one cyclic vector to the other. The unitary intertwines every distinguished source element and CP map.*

Proof. Faithfulness gives $\mathcal{N}_{\omega_s} = \{a : \omega_s(a^*a) = 0\} = \{0\}$. The GNS space is therefore the completion of $\mathcal{A}_s$ under $\langle a, b \rangle_s = \omega_s(a^*b)$, with cyclic vector $[1]$ and left representation $\pi_s(a)[b] = [ab]$. For two minimal cyclic realizations define

$$U\pi_1(a)\Omega_1 = \pi_2(a)\Omega_2.$$

Equality of the source state makes U isometric; cyclic minimality makes its range dense and hence U unitary. It intertwines the algebra representations, and Eq. (72) then makes it intertwine every distinguished element and CP map. Fixing the cyclic vector makes U unique on the cyclic span. $\square$

This proposition supplies an exact positive selection criterion. If the parent source law itself requires a completion C to realize one source-frozen signature Σ_s , and the induced signature map Φ_s is faithful modulo typed source isometry, then

$$\mathcal{D}_s(C) = \|\Phi_s(C) - \Sigma_s\|^2 \quad (73)$$

has exactly one zero completion class, provided an occupied zero exists and the source metric is positive definite after its declared null quotient. This proves uniqueness after complete source ownership; it does not prove that the current parent owns the required tuple.

Proposition 8 (Sector marginals do not select the joint source). *Complete faithful source states on the separate terminal sectors do not, in general, determine their joint source state.*

Proof. Take two two-level sectors with marginal states $I_2/2$. The faithful joint states

$$\rho_0 = \frac{I_4}{4}, \quad \rho_\varepsilon = \frac{1}{4} (I_4 + \varepsilon \sigma_z \otimes \sigma_z), \quad 0 < |\varepsilon| < 1, \quad (74)$$

have eigenvalues $1/4$ and $(1 \pm \varepsilon)/4$, respectively. Both partial traces equal $I_2/2$, while

$$\text{Tr}[\rho_0(\sigma_z \otimes \sigma_z)] = 0, \quad \text{Tr}[\rho_\varepsilon(\sigma_z \otimes \sigma_z)] = \varepsilon.$$

Thus identical complete marginals admit inequivalent faithful joint states. $\square$

Consequently the gravity–time–radiation, quantum–matter, and gauge assemblies cannot be promoted to one physically selected source merely by placing their marginal packets side by side. In finite dimension, a separating source signature must additionally fix cross-sector moments whose observables, together with all one-sector observables, span the self-adjoint joint source algebra modulo declared nulls. This is a finite information-rank condition. The current common action line and once-counted stress ledger establish compatibility and bookkeeping, but do not yet supply that full-rank cross-sector source family.

No-creation boundary. The GNS theorem represents every source-owned role but creates none. If record occurrence, hold ownership, the finite A8 connector, or microscopic continuum ownership is absent from $(\mathcal{A}_s, \omega_s, \mathbf{z}_s)$, representation uniqueness cannot supply it. Hence this section narrows the nature-selection blocker to one joint source-ownership and rank problem; it does not close that physical blocker.

10.12 Dimensional and claim-status audit

The principal quantities used by the completion have the following roles.

Quantity	Dimensions	Status
M_*, K_M	branch-dependent body coordinate and action Hessian	assumed stabilized; positive after true-null quotient
G_{SHR}	$L^3 M^{-1} T^{-2}$ in SI	not numerically predicted
Λ_{ren}	L^{-2}	renormalized infrared parameter; not predicted
$\mathcal{A}_X^{PC}, \mathcal{A}_*$	action	common units required
$x_O = \mathcal{A}_X^{PC} / \mathcal{A}_*$	dimensionless	body-action record ratio; A8b requires $0 < x_O \leq 2$ for distinct records
D_Q	dimensionless angle	adopted Fubini–Study/Bures distinguishability metric
ϵ_O	dimensionless	source-frozen Gram chord; $D_Q = 2 \arcsin(\epsilon_O/2)$
ΔH	energy	protocol-frozen energy spread
$\delta t_{\text{min}, O}$	time	conditional single-record lower bound
$\eta_{\text{rec}}, \tau_{\text{hold}}$	dimensionless probability, time	predeclared single-shot error ceiling and record-persistence interval

The resulting claim ledger is finite:

Statement	Status in this paper
Support-restricted common Uhlmann-amplitude bundle	derived once the occupied support and CP record instrument are supplied; physical occupation open
A8b mixed-state realization and $0 < x_O \leq 2$ range	derived consistency result; selection of $V_{r(x_O)}$ remains an explicit completion assumption
Record occurrence versus finite A8b	logically independent relative to the current source axioms; a physical QOR trial requires occurrence, A8b, and operational hold jointly
Finite GNS source signature	unique minimal cyclic finite completion up to typed unitary equivalence once the complete occupied source tuple is supplied
Separate terminal-sector marginals	insufficient to select the joint source; full-rank source-owned cross-sector moments remain open
Once-counted effective and quadratic ledger	derived once microscopic ownership and renormalization prescription are supplied; full microscopic Tau action open
Semiclassical Einstein/optics leaf	conditional standard-limit recovery
Trace-form Born leaf on fixed or classical-dynamic geometry	conditional on the state/effect valuation and covariant-transport assumptions
Born structure on a quantized metric	compatible only on a supplied constrained joint geometry-matter algebra; current construction open
Compatibility of classical limits	conditional consistency result
Pre-timing identity $x_A = x_Q$	Tau-specific conditional discriminator; physical source value and carrier occupation open
QOR inequality (33)	standard Bures-angle quantum speed limit after A8b; downstream consistency/falsification stage, not an extra Tau delay
Unique selection from the physical base and seed	exact finite criterion derived; current complete source tuple and cross-sector ownership not proved
Numerical constants, UV completion, full Standard Model	open
Empirical support for the Tau-specific common-source identification	open

11 Nontrivial representation condition

The parent/readout notation is not allowed to count as a Tau Core result by itself. Without an extra restriction, any effective theory could be written in the trivial form

$$\mathfrak{R}_\tau = \mathcal{O}, \quad \rho_\tau(s) = \mathcal{O}, \quad U = \text{id}_{\mathcal{O}}.$$

Such a representation is formally possible but scientifically empty.

Foundation Paper I therefore treats the following as a nontriviality requirement rather than as a theorem:

A promoted Tau Core branch must use a source-declared parent response and a readout family that cannot be reduced, without loss of the declared null, source, or known-limit structure, to the identity readout on the endpoint observable.

Equivalently, the parent structure must do at least one of the following before the endpoint is inspected: impose a quotient-safe null policy, relate two or more readouts through a shared source coordinate, recover a known physical limit, or exclude a wrong-family/control construction. Otherwise the branch is only a relabeling of the endpoint theory.

Known-limit recovery is not sufficient by itself. Many harmless rewritings of an existing theory can reproduce an already-known limit. Promotion therefore also requires at least one endpoint-blind consequence that could have failed: a frozen numerical value, a forbidden family, a null prediction, a source-side relation, or a control exclusion declared before endpoint inspection.

12 Nulls, quotients, and protected identity

Every physical readout has null directions: changes in the parent response that the readout treats as physically irrelevant. Some nulls are ordinary gauge redundancies. Some are observational limitations. Some are protected identity relations: distinctions that should not be counted as physical differences in a declared sector.

Let N_i denote the null relation for readout U_i . The quotient object is written

$$[r]_i \in \mathfrak{R}_\tau / N_i.$$

The readout is quotient-safe only if

$$r N_i r' \implies U_i(r) = U_i(r'). \quad (\text{QS})$$

Equivalently,

$$N_i \subseteq \sim_i, \quad U_i = \bar{U}_i \circ q_i, \quad q_i : \mathfrak{R}_\tau \rightarrow \mathfrak{R}_\tau / N_i,$$

for an induced quotient readout $\bar{U}_i : \mathfrak{R}_\tau / N_i \rightarrow \mathcal{O}_i$. If the quotient-safety condition (QS) fails, the readout assigns different observables to parent states that it also declares physically equivalent. Such a readout is internally inconsistent.

The converse is not required. A readout can identify more states than the declared null relation identifies. That is precisely readout loss. The important issue is whether the loss is allowed. In empirical branches this is handled by a source-freeze discipline: the readout-relevant source coordinates must be declared before endpoint scoring. In the foundation branch this becomes a source-priority problem: the selected readout should not erase protected source identity unless the erased distinction is explicitly routed to the null quotient.

A later formulation of the same idea is the readout-basin stability principle. For a declared 4D readout R_{4D} , a stable object-class O may be represented by the basin

$$\mathcal{B}_O = \{r \in \mathfrak{R}_\tau : R_{4D}(r) \simeq O\}.$$

Parent-response variation inside $\mathcal{B}_O$ preserves the observed object-class, while a visible transition or decay corresponds to crossing into another basin. In Foundation Paper I this is only a definition scaffold for object persistence under non-injective readout; it is not a proton-stability proof, a Standard Model derivation, or a decay-rate formula.

Definition 11 (Protected source identity). *A protected source identity is an equivalence relation Q_{src} on parent-side seeds or source representatives that the theory treats as source-level identity. A readout rule is source-faithful if it does not collapse distinct Q_{src} -classes into one selected source class without declaring the collapse as protected null structure.*

This definition is not an extra empirical claim. It is a consistency demand. Without it, a model can hide arbitrary distinctions in the parent space and recover them only after seeing an endpoint. That would make the theory unfalsifiable.

13 Endpoint blindness as a scientific rule

Endpoint blindness is the operational face of the foundation idea. A readout model is endpoint-blind when the source, kernel family, amplitude rule, and null policy are fixed before the target endpoint residual is inspected.

This is not a philosophical preference. It is the only way a broad readout framework can avoid becoming curve fitting. If a galaxy rotation residual, a cosmological distance residual, or a quantum outcome is used to choose the source class that then explains it, the model has leaked the answer into its input.

The minimal endpoint-blind record contains:

- (E1) a source object or source proxy;
- (E2) a declared readout family U_i ;
- (E3) a null and quotient policy;
- (E4) a source-overlap decision for all terminal readouts used together;
- (E5) a freeze artifact or equivalent timestamped declaration;
- (E6) forbidden endpoint fields;
- (E7) controls and failure conditions.

Foundation Paper I does not run such an endpoint test. It states why later branches need it. The philosophical content of Tau Core becomes scientific only when it produces source-frozen rules that can fail.

14 How effective readout-variation appears

Tau Core replaces primitive time evolution by readout-parametrized variation. Let $r = \rho_\tau(s)$ be atemporal at the parent-response level. Suppose a readout U_i produces a record that can be coordinatized by a parameter t_i :

$$U_i(r) = o_i(t_i).$$

Then the derivative

$$\frac{do_i}{dt_i}$$

is meaningful inside the readout. It need not correspond to a parent-time derivative. This is the minimal sense in which this paper uses the phrase “effective dynamics”: a parameterized variation internal to a readout, not a derived physical equation of motion.

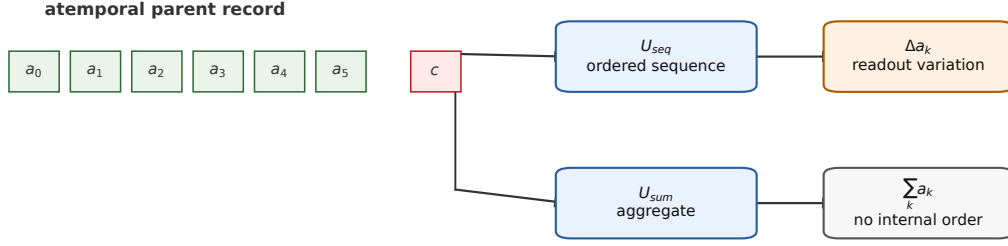
The distinction can be made sharper by considering two readouts of the same parent response. One readout may admit a temporal parameter t_A , while another admits a static relational record:

$$U_A(r) = o_A(t_A), \quad U_B(r) = C_B.$$

There is no contradiction. The first readout organizes the parent response as a time-indexed sequence. The second organizes it as a correlation or constraint. A conventional physical theory may then choose the first description because it is useful, local, and predictive. Tau Core’s claim is weaker: usefulness of a temporal parameter does not make time primitive.

This is also why observer time must be sector-typed. A clock used in a galaxy-rotation analysis, a lensing time-delay analysis, a cosmological distance analysis, and a quantum measurement protocol are not automatically the same parent object. They may share a local proper-time limit, but that limit must be derived or justified inside the declared branch.

F. Effective readout-time as an ordering



The parent record need not evolve for a readout to display an ordered time parameter.

Figure 8: A finite toy model for effective readout-time. The parent record is not itself evolving. The sequence readout U_{seq} exposes an ordered list and therefore supports finite differences Δa_k . The aggregate readout U_{sum} exposes a different observable and loses the internal ordering. Effective readout-variation is therefore readout-relative in this model.

15 Toy model with effective readout-time

The finite model above can be extended to show effective readout-time. Let

$$r = (a_0, a_1, \dots, a_n, c) \in \mathbb{R}^{n+2}.$$

Define a readout

$$U_{\text{seq}}(r) = (a_0, a_1, \dots, a_n),$$

and choose the index k as an effective readout-time coordinate. The readout sees a sequence a_k . A finite-difference law can be defined:

$$\Delta a_k = a_{k+1} - a_k.$$

Nothing in the parent object has evolved. The parent response already contains the indexed structure. The readout has supplied an ordering and therefore an effective readout-variation.

Now define a second readout

$$U_{\text{sum}}(r) = \sum_{k=0}^n a_k.$$

This readout sees no time sequence, only an aggregate. The hidden component relative to U_{sum} is the loss of ordered information. The hidden component relative to U_{seq} includes c . Again, hiddenness is readout-relative.

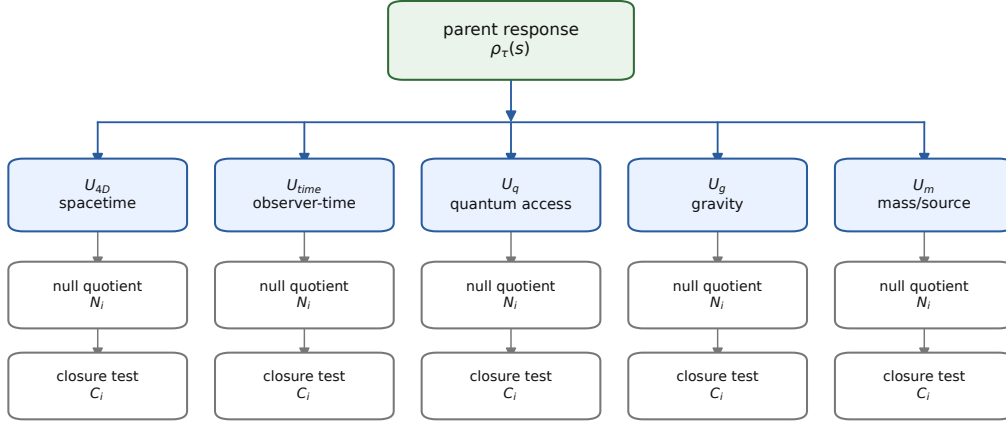
This nonphysical toy only makes “time is a readout” mathematically literal without backward causation or a second clock (Fig. 8).

Figure 9 locates the toy among the typed readout families.

16 Seven foundation postulates

The postulates below are not proved in this paper. They define the candidate framework. Later papers or theory notes may try to derive or weaken them.

C. Readout atlas with null and closure boundaries



Each sector needs its own codomain, null quotient, closure test, and validation boundary.

A shared parent response does not imply automatic additivity of readout components.

Figure 9: A parent response can have several sector readouts. The same parent object may be read as spacetime, observer-time, quantum access, gravitational response, or mass/source structure. Each sector requires its own codomain, null policy, closure test, and validation boundary. The paper does not assume that these readouts are automatically additive or mutually complete.

Postulate 1 (Tau substrate). *There exists a parent-level structure $\mathcal{T}$ that is not, at the foundation level, ordinary spacetime evolution.*

Postulate 2 (Seed-body response). *Physical configurations are represented at the parent level by endpoint-blind responses to seeds, stabilized as a morphology-conditioned body:*

$$s \in \mathcal{S}, \quad r = \rho_{\tau}(s) \in \mathfrak{R}_{\tau}, \quad r \longrightarrow M_{\tau}.$$

The generative direction is seed/base to body; the body is not constructed from a terminal residual or observer readout.

Postulate 3 (Endpoint blindness). *The response $\rho_{\tau}(s)$, and any source object used to select a readout kernel, must be specified without using the endpoint residual that the readout is later asked to explain.*

Postulate 4 (Readout emergence). *Observed physical descriptions are sector readouts:*

$$O_i = U_i^{M_{\tau}}(\rho_{\tau}(s)).$$

Time, spacetime, quantum access, gravity, and mass/source descriptions are readout candidates rather than primitive categories. The readout mediates the complete physical description, not merely a discrepancy relative to a chosen standard model.

Postulate 5 (Effective readout variation). *Time evolution appears only when a readout admits a stable ordering or clock coordinate. A derivative such as*

$$\frac{d}{dt}U_i(\rho_{\tau}(s))$$

is an effective readout derivative, not a primitive parent-time derivative and not, by itself, a physical law of motion.

G. Claim-boundary ladder

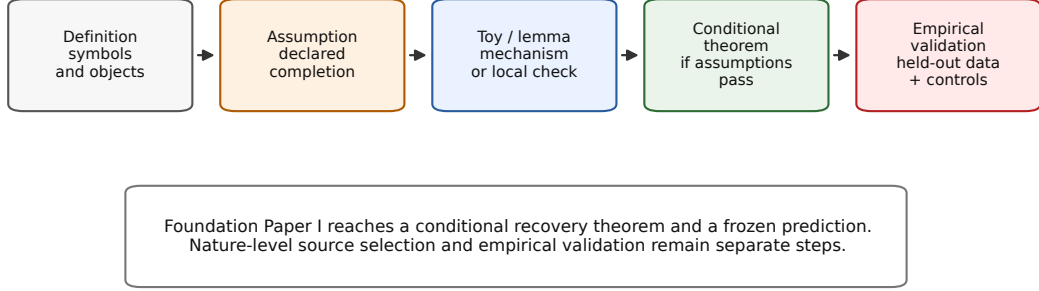


Figure 10: The claim ladder for Foundation Paper I. The present manuscript reaches conditional standard-limit results and one frozen discriminator, but not nature-level source selection or empirical validation.

Postulate 6 (Hidden defect). *If a readout and its allowed relift fail to recover the parent-response representative modulo the declared null relation, the obstruction $\omega_i(r) \neq 0$ is recorded as a hidden defect of that readout. A coordinate difference $J_i U_i(r) - r$ is only a chart-level representative of that obstruction.*

Postulate 7 (Tension as readout failure). *Some observational or theoretical tensions may be modeled as readout, closure, or relift defects. This is a research hypothesis. It becomes physical only after source-frozen rules, controls, known limits, and held-out tests are supplied.*

17 The claim ladder

Tau Core separates definitions, postulates, conjectures, conditional theorems, toy models, numerical evidence, and validation; they are not interchangeable (Fig. 10).

The most common failure mode for broad foundation programs is promotion by language: a metaphor becomes a mechanism, a mechanism becomes evidence, and evidence becomes validation. This paper avoids that promotion. For example, the conditional clock and Born constructions do not prove their physical source occupation. Saying that dark-sector language may be a hidden-defect language does not remove dark matter or dark energy.

Figure 11 gives the same discipline in compact form. Green entries indicate definitions or conditional results supplied here; orange and red entries indicate explicit non-claims and blockers. The point is not to make the paper look complete, but to make incompleteness visible.

18 A compact finite toy witness

This deliberately nonphysical witness records only the quotient logic. Let $\mathfrak{R}_\tau = \mathbb{R}^3$, $r = (a, b, c)$, and

$$U_A(r) = (a, b), \quad U_B(r) = (a, c), \quad J_A(x, y) = (x, y, 0), \quad J_B(x, z) = (x, 0, z).$$

In this linear chart,

$$h_A(r) = (0, 0, -c), \quad h_B(r) = (0, -b, 0).$$

H. Foundation Paper I claim-status matrix

	Defined here	MVP assumption	Conditional result	Tau empirical test	Source selection
Weak Tau spine	specified				open
Stabilized body	specified	specified	conditional		open
Observer descent	specified	specified	conditional		open
GR / quantum limits	specified	specified	conditional		open
QOR prediction	specified	specified	conditional	open	open
Nature-level theory				open	open

Green: specified inputs. Orange: conditional results. Red: open physical selection or empirical status.

Figure 11: Claim-status matrix for Foundation Paper I. The paper defines and postulates a parent/body/readout architecture, adopts one conditional completion, recovers standard leaves under explicit assumptions, and states the QOR discriminator. Nature-level selection and empirical validation remain open.

Thus a coordinate hidden under one readout can be visible under another. For U_A , the quotient-safe null relation fixes a, b and ignores c . Changing the readout afterwards to $U_A^{\text{bad}}(a, b, c) = (a, b + \alpha c)$ while retaining that null relation is inconsistent: equivalent representatives would yield different observables. Likewise, a source rule $c = f_{\text{src}}(a, b; \theta)$ is admissible only when f_{src} and θ are frozen before endpoint inspection.

The witness proves no dark-sector, quantum, or gravitational mechanism. It only shows that hiddenness is readout-relative and that null policies and source rules cannot be changed to rescue an endpoint.

19 Morphological body versus projected morphology

The phrase “morphology” can be misleading if left unqualified. In ordinary astronomical use it often means visible shape or catalogue type. In Tau Core, the primary object is not the observed shape. It is the Tau-side morphological body M_τ .

The morphological body is the structured source object that conditions readouts. A galaxy image, a rotation curve, a cosmic-web tensor, a lensing map, or a quantum measurement context can be a projected handle on the body, but not the whole body. This distinction became necessary because a single 4D appearance can underdetermine the readout-relevant source. For example, a present projected shape may not contain enough information about global incidence/order, observer/path geometry, closure class, or null structure.

For Foundation Paper I, the minimal role of M_τ is:

$$O_i = U_i^{M_\tau}(\rho_\tau(s)). \quad (75)$$

The superscript indicates that the readout map is selected or conditioned by the morphological body. Equivalently one may regard U_i as a map on $\mathcal{M}_\tau \times \mathfrak{R}_\tau$. This does not say that M_τ has been physically derived. It says that readout selection is not source-free.

For a concrete branch, M_τ must also be admissible and frozen. This means that the projected handles used to select M_τ , the allowed family of readout kernels, and any source-overlap decisions between terminal readouts must be declared before endpoint scoring. If M_τ is changed

after seeing the endpoint residual, the branch is phenomenological-only unless an independent source review reopens the selection.

This point also prevents double counting. If two proposed correction terms use the same source coordinate, they should not be added independently unless a source-side split justifies the separation. Otherwise the readout model is only relabeling the same source information twice.

20 Observer time

Tau Core treats observer time as a readout, not as a primitive parameter on the parent side. The schematic form is

$$t_{\text{obs}}^S = U_{\text{time}}^S(\rho_\tau(s), M_\tau, \mathcal{O}_S, \gamma_S), \quad (76)$$

where S is a sector, $\mathcal{O}_S$ is observer-interface data, and γ_S denotes path or section data relevant to that sector.

Equation (76) is the weak-spine discipline rule. The MVP completion adds an oriented clock descent and the conditional resolution law (33); it still does not derive physical occupation of that clock carrier. Proper-time recovery, any non-relativistic observer correction, and the operational measurement of such a correction must be checked without circularity in the relevant branch.

This also clarifies why future-directed or history-like terms in Tau Core should not be read as backward causation. They are components of a deeper parent response whose different 4D readout slices may appear as past, present, or future. The parent object is not evolving in an external time. The observed ordering is created inside the readout.

21 Quantum readout

The weak-spine statement is simply $q = U_q^{M_\tau}(\rho_\tau(s))$, where q is an access-state or probability-assignment record. That notation alone does not imply Hilbert space, interference, or the Born rule. Existing reconstruction programs show how much operational, convex, symmetry, and composition structure is normally required [18, 9, 27, 5].

The MVP branch makes a finite stronger commitment: it adopts the positive state/effect completion A6 and therefore obtains the trace rule (24). The mathematical implication from the valuation assumptions to trace form is a conditional representation result. What is not derived is the physical Tau-parent selection of those valuation assumptions. This distinction is essential: information loss or finite readout capacity by itself gives a coarse graining, not specifically complex quantum mechanics.

Quantum and gravity are coupled in the MVP because $\langle T_{\mu\nu}^q \rangle_{\text{ren}}$ is the metric derivative of the same matter functional that defines quantum probabilities. The metric may be kept classical at the semiclassical leaf, while its second variation defines a stress-noise kernel and a stochastic-gravity refinement. The paper does not claim a nonperturbative metric-quantization theorem, a unique tensor-product structure, interacting ultraviolet QFT, or a complete measurement ontology. It claims the narrower common-functional and compatible-limit results of Sec. 10.

22 Gravity and dark-sector readouts

The weak gravity branch motivates the hidden-defect vocabulary. A gravitational readout may fail to relift all source-relevant structure into a Newtonian, relativistic, or closure-compatible description. In abstract notation:

$$\omega_g(r) = 0 \quad \Longleftrightarrow \quad J_g U_g(r) N_g r, \quad r = \rho_\tau(s). \quad (77)$$

If $\omega_g(r) \neq 0$ and the obstruction record survives the declared null quotient with a measurable 4D readout, it becomes a candidate residual diagnostic. A coordinate difference

$$h_g = J_g U_g(r) - r$$

may be used only after choosing a linearized gravitational chart in which the subtraction is meaningful. This is not the basis of the MVP gravitational law: the full gravitational readout is instead generated by the common functional and has the conditional Einstein leaf (21). Empirical residuals are downstream model comparisons and must not be used to choose the body or completion.

The same discipline applies to dark-sector language. Dark-matter- or dark-energy-like inference could, in principle, contain body morphology, observer descent, omitted conventional sources, calibration, or genuinely new degrees of freedom. Foundation Paper I does not choose among them. To make a dark-sector claim one would need a source-frozen forward construction, conventional baseline comparisons, covariance and systematics treatment, and an outcome that distinguishes these alternatives. The recovery of Einstein gravity does not, by itself, explain the dark sector.

23 Cosmological readout

Cosmology is another natural readout target because the observed expansion history, distance ladder, growth data, lensing data, and CMB inference are all compressed through model-dependent readout assumptions. A Tau Core cosmological readout might be written

$$O_{\text{cosmo}} = U_{\text{cosmo}}(\rho_\tau(s), M_\tau^U, \gamma_{\text{path}}, \mathcal{O}_{\text{obs}}), \quad (78)$$

where M_τ^U is a universe-scale morphological body, γ_{path} is path/environment data, and $\mathcal{O}_{\text{obs}}$ is observer-interface data.

Again, the notation is not a cosmological result. It is a warning against compressing all cosmological tensions into a single function of the scale factor. A real cosmological Tau Core branch would need multi-probe covariance, source-native void/web/path/time descriptors, and a clear relation to Λ CDM limits. Neither Λ_{ren} in Eq. (21) nor its numerical value is derived here.

24 Position relative to MOND and dark-matter programs

Tau Core should not be framed as anti-MOND or anti-dark-matter. A safer and more accurate position is that it tries to supply a third-level language in which MOND-like phenomenology and dark-matter-like signals may both be effective readouts of deeper source structure.

For the MOND program, the attractive point is that Tau Core does not begin by assuming that the standard cold-particle halo picture is the whole story. It allows the possibility that galaxy-scale acceleration regularities are effective consequences of a deeper gravitational/readout structure. The limiting point, from a strict MOND perspective, is that Tau Core does not promote MOND itself to final ontology. It treats MOND-like laws, if they appear, as possible intermediate effective rules:

$$\text{MOND-like regularity} \quad \rightsquigarrow \quad \text{effective readout limit of a deeper morphology.}$$

This is not a MOND derivation in this paper. It is a claim-boundary statement: Tau Core may try to explain why a MOND-like scale or relation appears, but it must still derive the scale, recover known limits, and pass controls.

For the dark-matter program, the caution is different. Tau Core should not say prematurely that dark matter is an illusion. Lensing, CMB, BAO, cluster, and structure-growth data provide

strong evidence for extra gravitational structure in the standard cosmological inference pipeline. A Tau Core branch would have to ask a more decomposed question:

$$\begin{aligned} \text{dark-sector signal} \sim & \text{gravitating content} + \text{baryonic remnants} \\ & + \text{modified response} + \text{readout/path/morphology effects}, \end{aligned}$$

where the equality is schematic, not a model equation. The scientific task is to determine which components are present in which terminal readout, and which component, if any, behaves like collisionless matter.

Thus the intended public framing is:

Tau Core is neither anti-MOND nor anti-dark-matter. It is a readout-level formal language in which MOND-like acceleration laws, dark-matter-like lensing/cosmological signals, and observer-dependent residuals can be separated into source-frozen components and tested against one another.

This framing avoids a victory narrative. It does not say that existing communities were simply wrong. It says that different communities may have identified different projections of a deeper problem, and that the useful question is how to split those projections without endpoint leakage.

25 Comparison with adjacent ideas

Tau Core overlaps in spirit with several families of ideas, but it should not be confused with them.

Relational time. Relational-time approaches show how dynamics can be recovered through correlations. Tau Core agrees that external time need not be primitive, but it places this inside a broader readout atlas. Time is one readout among several, not the only reconstruction target.

Shape dynamics and relational configuration. Shape-dynamics-inspired work emphasizes that spacetime structure may be reformulated through different symmetry or relational choices [16]. Tau Core uses this as a cautionary analogy: one must state which structure is primitive and which is recovered. It does not import shape dynamics as a proof.

Quantum reconstruction. Quantum reconstruction programs provide a model for turning operational principles into formal structure. Tau Core’s quantum branch must meet a similar standard. It cannot derive Hilbert space merely by naming a quantum readout. The adopted effect valuation in Sec. 10.5 is an explicit conditional completion of that gap, not a derivation from the weak Tau spine.

Semiclassical and stochastic gravity. The coupled equation (21) belongs to the standard semiclassical-gravity family, and its metric second variation is naturally related to stress fluctuations and stochastic gravity [22]. Tau Core does not claim those equations as new. Its candidate contribution is the body-first common ownership of gravity, quantum effects, and observer records.

Locally covariant quantum field theory. Local covariance, microlocal/Hadamard regularity, and renormalized local observables provide the appropriate standard for a QFT readout on curved spacetime [7, 20, 21]. The MVP assumes this infrared structure. It does not reconstruct the full local net or interacting theory from parent morphology.

Quantum speed limits. The QOR inequality has the same inverse-energy-spread dimension as Mandelstam–Tamm and geometric quantum-speed-limit relations [25, 3, 36]. The proposed discriminator is therefore not the inverse-energy bound or the standard Gram-chord/Bures-angle conversion. It is the source-frozen physical identification of the morphological factor ϵ_O with the occupied observer carrier’s Uhlmann-amplitude distance.

Holography and error correction. Holographic quantum error correction shows that bulk-like structure can be encoded and recovered nontrivially from boundary data [2]. Tau Core treats this as an analogy for readout/reconstruction discipline, not as evidence for the Tau ontology.

26 What would count as progress

The framework becomes stronger only if abstract readout language is converted into bounded tasks. Examples include:

- (1) deriving a non-circular parent response law ρ_τ ;
- (2) defining a source-realizable morphological body M_τ ;
- (3) proving that a readout map has a quotient-safe null policy;
- (4) deriving a known physical limit, such as proper time or Newtonian gravity, as a controlled readout limit;
- (5) producing source-frozen empirical rules that survive held-out tests;
- (6) preserving negative results by demoting failed kernels instead of retuning them after end-point inspection.

The framework has now passed the earlier milestone of constructing one worked completion: Sec. 10 fixes a variational response type, recovers two standard leaves, and freezes one discriminator. The next milestone is not another general layer or another terminal. It is to decide the physical selection question in Eq. (70), preferably by deriving the occupied completion from a deeper source law and independently by testing Eq. (33).

The weakest useful form of progress is a toy model that exposes a blocker. A failed theorem can be valuable if it shows which extra principle is needed. For example, local convexity or a positive Hessian may not force global exposed-face duality; finite contextual grammars may not force Hilbert space; and source stability may not force a unique source basis. Such negative results help keep the framework honest.

The weak Tau spine still permits several mathematical types for a parent response law. Candidate types include:

variational law	ρ_τ selected by an action or extremal principle,
constraint law	ρ_τ selected by admissibility or closure equations,
algebraic closure	ρ_τ fixed by source/readout consistency,
probabilistic kernel	ρ_τ given by an endpoint-blind transition kernel,
fixed point	ρ_τ selected as a stable solution of a response operator,
functorial rule	ρ_τ induced by structure-preserving source maps.

The MVP makes one finite choice—an ordered variational completion—without claiming that all Tau branches must use it. Any alternative branch must still declare its type before claiming a physical law.

The largest remaining weakness is therefore not the absence of every worked chain. It is the missing implication from the physical source to the chosen completion. The relevant standard is now

$$\text{source} \rightarrow \rho_\tau \rightarrow U_i^{M_\tau} \rightarrow \text{known limit or frozen falsifiable consequence}$$

without endpoint leakage, together with a proof that the physical source actually occupies the response law used in that chain.

27 What would count as failure

A reviewer should not have to infer the failure conditions. At the foundation level, Tau Core would be weakened or demoted if the following failure modes persist after reasonable attempts to repair definitions:

- (F1) **No source-realizable parent object.** If no non-circular parent-side source class can be specified for a branch, the branch remains vocabulary rather than physics.
- (F2) **No quotient-safe readout.** If declared null-equivalent parent representatives produce different observables, the readout is internally inconsistent.
- (F3) **No known-limit recovery.** If a branch cannot recover the relevant standard limit, such as proper time, Newtonian gravity, Born probabilities, or Λ CDM behavior where appropriate, it should not be promoted.
- (F4) **Endpoint leakage.** If the source class, kernel family, or amplitude rule is selected using the residual it later explains, the result is a fit, not a Tau Core derivation.
- (F5) **Uncontrolled rescue.** If every failure is repaired by adding an unconstrained terminal after endpoint inspection, the framework loses scientific content.

These are not merely methodological preferences. They are the conditions under which the present foundation language can fail to become a physical research program. A negative result satisfying these conditions should be preserved as a real demotion, not hidden by changing terminology.

For later branches, demotion should be recorded explicitly. The minimal status labels have the following strict meanings:

Status	Strict meaning
TOY-ONLY	no physical codomain, units, or operational observable supplied
PHENOMENOLOGICAL-ONLY	fits or organizes known data but lacks parent-source derivation
BLOCKED	missing known-limit recovery, quotient-safety, or source-realizability
REJECTED	frozen prediction fails holdout/control, or endpoint leakage is found

Forbidden rescue operations include post-hoc kernel exchange, endpoint-selected morphological parameters, retrofitted path factors, and covariance or null policy changes made after endpoint inspection. A promoted branch must provide at least one frozen numerical value, relation, exclusion, or null prediction that could have failed.

Minimal failed-branch example. Let a toy branch declare a null policy $n \sim n'$ and a readout

$$R(x, n) = x.$$

If the branch later adds a correction An to explain an endpoint residual, then it has used a declared null coordinate as a physical observable. The branch fails quotient safety. If A or the choice of n -dependence was selected after inspecting the endpoint residual, it also fails endpoint blindness. Such a branch is not a Tau Core derivation; it is demoted to REJECTED or, if no endpoint was used, at best PHENOMENOLOGICAL-ONLY. This example is deliberately trivial: its purpose is to show that the framework can say “no” before any exotic physics is invoked.

28 Likely objections

A foundation framework should be written so that criticism has handles. We therefore list several likely objections and the intended answer.

Objection 1: this is only a change of language. This objection is serious. The framework is only useful if readout language forces new constraints. The minimum non-linguistic constraints are endpoint blindness, null/quotient safety, source-overlap discipline, known-limit recovery, and explicit failure conditions. Without these, Tau Core would be a vocabulary rather than a research program.

Objection 2: the parent response is unobservable. The parent response is not directly observed by assumption. But this alone is not fatal; many theoretical objects are inferred through constrained effects. The relevant question is whether a declared readout rule produces predictions or exclusions that could have failed. If no such rule is produced, the parent response remains a metaphysical placeholder.

Objection 3: the framework can explain anything. This is the most important danger. Tau Core must therefore preserve negative results. A failed frozen kernel should be demoted, not retuned after looking at the residual. A wrong readout family should remain wrong. A source class that cannot be independently justified should not be promoted. If the framework always rescues itself by adding another projection, it fails as science.

Objection 4: atemporal response sounds like a block universe. The difference is the block-readout relation (3). A block universe treats a 4D history as the final object. Tau Core treats a 4D history as one possible readout. The distinction matters only if other readouts impose additional source-frozen constraints. If they do not, Tau Core collapses back into ordinary effective description.

Objection 5: the displayed physical laws are assumed rather than derived from the Tau parent. Partly correct. Within the frozen completion, the terminal equations follow from one variational packet and their compatibility is nontrivial. But the completion contains explicit Lorentzian, state/effect, and observer-carrier assumptions, and the physical base-seed source has not been proved to select them. The manuscript therefore claims conditional recovery, not a first-principles derivation of GR or quantum theory from the minimal Tau base.

Objection 6: the QOR inequality is only a quantum speed limit. The $1/\Delta H$ dependence is indeed not new. The only proposed new content is the independently reconstructed same-source identity $x_A = x_Q$, equivalently the identification of the body ratio with the Uhlmann chord. The conversion from that chord to the Fubini–Study/Bures record angle is

standard and is proved inside the common-Gram completion, not by positivity of the weak Tau spine alone. The finite carrier prototype closes the mathematical and uncertainty bookkeeping but does not yet supply a physically occupied common-Gram carrier. If ϵ_O is fitted from the timing endpoint, the prediction is circular. A stable one-shot record below the bound falsifies the completion; absence of a threshold ordering is decisive only when bound saturation or equal slack has been established independently.

Objection 7: the common functional may hide double counting. Equations (12)–(16) now supply the term-by-term effective ledger. Retained body stress is varied only from $S_{\text{body}}^{\text{ret}}$; integrated quantum/apparatus stress and effect responses come only from W_O^{ren} ; local counterterms occupy one side of the renormalization convention; and Green responses are inverse second variations, not extra summands. Mixed blocks are handled by the exact Schur reduction (15). This closes the effective and quadratic bookkeeping objection once microscopic ownership is supplied. It does not derive that ownership partition or a full nonlinear microscopic Tau action.

Objection 8: fluctuating geometry may invalidate the Born leaf. Correct. Equation (24) is established only on the fixed descended metric under A6. If effects or the carrier Hilbert structure vary with unresolved body/metric fluctuations, positivity, completeness, and noncontextual additivity must be proved on the joint algebra. The present paper supplies no such nonperturbative theorem.

Objection 9: semiclassical recovery may fail in extreme regimes. The displayed gravity leaf does not establish stability through black-hole evaporation, Page-curve regimes, topology change, or large stress fluctuations. Such failures would delimit or reject this MVP completion; they cannot be counted as evidence for a hidden Tau correction without a separate source-frozen derivation.

29 Structural predictions rather than anomaly corrections

The ontological question is whether the Tau Core language can ever yield a prediction that is not merely a correction to an anomaly in an existing framework such as MOND, Λ CDM, GR, or quantum theory. The answer in Foundation Paper I is conditional.

A Tau Core prediction is not counted as structurally original merely because it adds a residual term to a baseline equation. It would count only if a source-frozen parent/morphology rule implies a constraint before the target endpoint is inspected. The schematic form is

$$\text{Struct}_\tau(s, M_\tau) = 0 \implies \text{Constraint}(U_i^{M_\tau}, U_j^{M_\tau}, N_i, N_j),$$

or an exclusion:

$$\text{Struct}_\tau(s, M_\tau) \implies \text{readout pattern } P \text{ is inadmissible.}$$

Examples of the relevant kind of constraint include:

- a compatibility condition forcing two readouts to co-vary;
- a null/quotient rule excluding an apparent correction as double counting;
- a basin-transition invariant restricting which readout transitions can appear as stable 4D events;
- a rank, connectivity, or frame-gluing condition that must hold before a Hilbert-like or metric-like representation is allowed;

- a frozen source rule that predicts a missing, forbidden, or correlated feature in a holdout readout.

Such a prediction is Tau Core-specific only if it is fixed from the parent source, morphological body, null policy, and readout compatibility rules before endpoint inspection. If the same structure is chosen because a known baseline leaves a residual, it is an anomaly-motivated correction rather than a parent-morphological prediction. Equation (33) is the first conditional example in this paper: its morphology-dependent factor is frozen upstream, while its role in quantum-state record distance is derived from the explicit common-Gram completion identity. It is not yet a validated physical example because common-Gram carrier occupation, laboratory realization, and the empirical test remain open.

A concrete exclusion at framework level. The framework does forbid one nontrivial modeling move. Suppose two proposed sector corrections are written as

$$\Delta O_i = A_i q, \quad \Delta O_j = A_j q,$$

where q is the same source-side coordinate or proxy. Unless the branch declares a source-side split $q = (q_i, q_j)$, an orthogonality/covariance rule, or a shared-parent covariance model before endpoint inspection, the two terms cannot be treated as independent evidence or added as independent corrections. They are the same source degree of freedom read twice. Thus Tau Core forbids “same-source double counting” as an admissible explanation. This is not a new physical prediction, but it is a structural exclusion that goes beyond renaming a residual.

30 Minimal mathematical maturity checklist

Before a branch of Tau Core should be treated as more than a toy model, it should satisfy at least the following checklist.

Requirement	Meaning
Source declaration	The parent-side source or proxy is declared before endpoint scoring.
Body-first order	The stabilized body is solved and frozen before terminal variation.
Common ownership	Shared stresses, currents, and probes are varied from one once-counted functional.
Readout codomain	The observable space $\mathcal{O}_i$ is specified.
Null policy	Gauge, redundancy, and protected-null directions are declared.
Quotient safety	Equivalent parent representatives give the same readout.
Relift policy	The branch states whether and how J_i is defined.
Known limit	The readout recovers a standard limit where required.
Dimensional audit	Common units and dimensionless ratios are identified without hidden calibration.
New discriminator	At least one upstream-frozen consequence could differ from standard recovery alone.
Failure rule	Failed controls cause demotion, not post-hoc retuning.

This checklist is intentionally weaker than a full physical theory. It is also stronger than a metaphor. A branch that cannot fill these rows should not be advertised as a physical result.

31 Next decisive work

The next foundation-stage calculation should not add another general framework layer or another unconstrained terminal. It should close or sharply reject the open first arrow in

$$B_\tau^{\text{phys}} + s_U \longrightarrow \Gamma_{\text{MVP}} + \text{state/effect completion} + \text{observer carrier}.$$

This requires either a source theorem selecting the full occupied completion or a source-frozen test that rejects it. The source theorem now has a finite target: a source-owned finite algebra/state/role packet whose mixed cross-sector moments have full joint information rank, including occupation of the support/record instrument in A8a-OCC, selection of the same-action connector $V_{r(x_O)}$ in A8b, affiliation of the global Dirac operator with the recovery algebra as in Proposition 5, and one microscopic ownership partition whose integration yields Eq. (14). Proposition 7 would then make the finite completion unique automatically. Deeper quantum composition, interacting QFT, dark-sector interpretation, cosmological realization, and particle-sector structure remain downstream tasks; none is imported here as support for the present completion.

Proposition 4 prevents either of the first two targets from being used as proof of the other. A successful source theorem may select them jointly, but must contain an explicit coupling law rather than infer occurrence from metric agreement or metric agreement from a record event. It must also supply, or leave independently testable, the single-record retention condition.

A first laboratory worked model should be chosen from a carrier with an independently controllable two-record manifold, state tomography, calibrated energy spread, and a quasistatic work or phase response. Superconducting qubits, trapped-ion or spin-1/2 controls, two metastable atomic levels, and photon-polarization records are candidate platforms, not asserted Tau carriers. Promotion of any one candidate requires source-side definitions of q , $M(q)$, $\mathcal{A}_{\text{cal}}$, $\rho_{O,0}$, $\rho_{O,1}$, and ΔH , followed by the two pre-timing tests above.

32 A compact formal summary

The weak framework is the object package

$$\mathfrak{T} = (\mathcal{S}, \mathfrak{R}_\tau, \rho_\tau, M_\tau, \{U_i\}, \{N_i\}, \{J_i\}).$$

Here $\mathcal{S}$ is a seed/source domain, $\mathfrak{R}_\tau$ is a response domain, ρ_τ is the endpoint-blind response map or relation, M_τ is the morphological body, U_i are readouts, N_i are null policies, and J_i are optional relifts.

The minimum-viable completion augments it to

$$\mathfrak{T}_{\text{MVP}} = (\mathfrak{T}, M_*, K_M, \Gamma_{\text{MVP}}, \text{Desc}_O, \rho_O, \{F_{O,a}\}, \epsilon_O). \quad (79)$$

The minimal consistency conditions are

$$s \text{ is fixed without endpoint leakage,} \quad (80)$$

$$r = \rho_\tau(s) \text{ and } M_* \text{ are defined on the declared source domain,} \quad (81)$$

$$D_M \mathcal{A}_{\text{body}}(M_*) = 0, \quad K_M > 0 \text{ after the true-null quotient,} \quad (82)$$

$$\text{the body is frozen before terminal variations,} \quad (83)$$

$$r N_i r' \Rightarrow U_i(r) = U_i(r'), \quad (84)$$

$$\omega_i(r) = 0 \iff J_i U_i(r) N_i r, \quad (85)$$

$$h_i(r) = J_i U_i(r) - r \text{ is only a chart-level representative if defined,} \quad (86)$$

$$\text{shared terminals are extracted once from } \Gamma_{\text{MVP}}, \quad (87)$$

$$\text{known limits and the QOR test are frozen before physical promotion.} \quad (88)$$

The strongest positive claim is that $\mathfrak{T}_{\text{MVP}}$ is an internally composable conditional theory with the standard leaves (21) and (24) and the discriminator (33). The strongest negative claim is Eq. (70): the current physical source reduct does not uniquely select that completion.

33 What is not proved

For clarity, we list the main non-results.

- The paper does not prove that the Tau substrate, universal seed, or complete universe-scale morphological body exists physically.
- It does not derive the adopted enriched body action and completion uniquely from the physical base-seed source.
- It does not derive the microscopic ownership partition or full nonlinear action whose integration yields the once-counted effective functional in Eq. (14).
- It does not predict G , Λ , $\hbar$, gauge couplings, particle masses, or other constants numerically from parent data.
- Conditional Einstein and Born recovery is not an unconditional derivation of GR or quantum mechanics from the minimal Tau base.
- It does not provide ultraviolet quantum gravity, nonperturbative interacting QFT, a complete measurement ontology, or a metric-quantization theorem.
- It does not derive the full Standard Model, its spectrum, generations, mixing, or neutrino sector.
- The QOR inequality is not a universal time atom or an extra delay. The support theorem and mixed-state realization prove that the common-Gram completion is mathematically nonempty, but the physical parent has not been shown to occupy the record instrument or select $V_{\text{phys}} = V_{r(x_O)}$. No laboratory carrier has validated the same-source identity $x_A = x_Q$.
- It does not eliminate or explain dark matter or dark energy.
- No empirical result presented in this manuscript validates the Tau ontology.

These limitations are not footnotes. They are part of the claim. A foundation paper that hides its blockers is not useful. The goal is to make the blockers visible enough that later work can attack them one at a time.

34 Conclusion

Tau Core is presented as an atemporal morphological readout framework with one frozen conditional completion. The weak proposal is that observed physical descriptions are body-conditioned readouts of an endpoint-blind parent response:

$$s \mapsto \rho_\tau(s) \mapsto U_i^{M_\tau}(\rho_\tau(s)).$$

In this reading, a 4D spacetime block is not the final ontology; it is one readout:

$$M_{4D} = U_{4D}^{M_\tau}(\rho_\tau(s)).$$

Physical time is an oriented observer descent of body-resident order, not the meaning of τ itself and not a separate substance between parent and observer.

The paper's main advance over a framework-only statement is the finite chain (7). A body is stabilized first; one once-counted functional then supplies gravity, matter/gauge, quantum-effect, and clock variations. Under explicit assumptions this gives compatible semiclassical-Einstein and Schrödinger–Born leaves. It also gives the conditional QOR inequality (33). The known inverse-energy dependence is not the proposed novelty. The proposed new content is the independently testable same-source identity $x_A = x_Q$. The common-Gram minimum-lift identity then proves how this factor determines the required Fubini–Study/Bures record angle. The QOR timing inequality is the resulting standard quantum speed limit, not an additional Tau delay. Physical record occurrence, selection of the $V_{r(x_O)}$ connector, and operational retention are three jointly necessary conditions. Proposition 4 proves, relative to the present source axioms, that occurrence and finite A8b do not select one another. A deeper source law may still select both, but that common physical selection remains part of the completion problem. Proposition 5 supplies an exact sufficient route: an affiliated global Dirac operator is reduced by the central recovery blocks, making the finite spectral action additive. The current parent has not yet selected that affiliation law. Proposition 7 now proves that one complete occupied finite source algebra/state/role packet would select the finite completion up to typed unitary equivalence. Proposition 8 proves why the currently assembled sector marginals are insufficient: source-owned cross-sector moments of full joint information rank are still required.

The strongest honest summary is therefore:

Tau Core now has one auditable conditional route from an atemporal morphological body to compatible gravity and quantum standard limits plus a candidate observer-resolution discriminator. It does not yet have a proof that the physical parent owns the complete joint source signature selecting and occupying this common-Gram route.

This distinction supplies a clean decision point. A deeper source theorem or a successful source-frozen QOR test would promote the completion. Failure of a reliable single-record event below its source-frozen QOR bound would reject this completion. Adding unconstrained terminals after such a failure would not be a repair; it would abandon the minimum-viable theory tested here.

A Mixed-state distance and QFI conventions

To avoid the common squared-versus-root fidelity ambiguity, this manuscript uses the *squared* Uhlmann fidelity

$$F(\rho, \sigma) = \left[\text{Tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right]^2, \quad 0 \leq F \leq 1. \quad (89)$$

The root fidelity is $f(\rho, \sigma) = \sqrt{F(\rho, \sigma)}$. The Bures angle used in Eq. (29) is therefore

$$D_B(\rho, \sigma) = \arccos f(\rho, \sigma) = \arccos \sqrt{F(\rho, \sigma)}. \quad (90)$$

For pure states, $F = |\langle \psi | \varphi \rangle|^2$, so $D_B = \arccos |\langle \psi | \varphi \rangle|$, the Fubini–Study angle.

For a differentiable one-parameter mixed-state path, the infinitesimal Bures line element and quantum Fisher information $\mathcal{F}_Q$ satisfy

$$ds_B^2 = \frac{1}{4} \mathcal{F}_Q(t) dt^2. \quad (91)$$

For unitary evolution generated by a self-adjoint H ,

$$\mathcal{F}_Q(t) \leq \frac{4\Delta H(t)^2}{\hbar^2}, \quad \frac{ds_B}{dt} \leq \frac{\Delta H(t)}{\hbar}, \quad (92)$$

which integrates to Eq. (32) [36]. A nonunitary channel must instead use its actual QFI path speed; Eq. (92) is not assumed for it.

The common-Gram completion fixes the notation

$$x_O = \frac{\mathcal{A}_X^{PC}}{\mathcal{A}_*} = \epsilon_O^2 = 2(1 - \sqrt{F}), \quad D_Q = 2 \arcsin(\epsilon_O/2).$$

Hence the linear inequality $D_Q \geq \epsilon_O$ is a conservative corollary, with equality only in the zero-separation limit. The exact state-distance quantum speed limit is the first inequality in Eq. (33).

References

- [1] A. Fabricio Albuquerque, Fabien Alet, Clement Sire, and Sylvain Capponi. Quantum critical scaling of fidelity susceptibility. *Physical Review B*, 81:064418, 2010.
- [2] Ahmed Almheiri, Xi Dong, and Daniel Harlow. Bulk locality and quantum error correction in AdS/CFT. *Journal of High Energy Physics*, 2015(4):163, 2015.
- [3] Jeeva Anandan and Yakir Aharonov. Geometry of quantum evolution. *Physical Review Letters*, 65:1697–1700, 1990.
- [4] Julian Barbour. *The End of Time: The Next Revolution in Physics*. Oxford University Press, 1999.
- [5] Howard Barnum and Alexander Wilce. Post-classical probability theory. *arXiv:1205.3833*, 2014.
- [6] Luca Bombelli, Joohan Lee, David Meyer, and Rafael D. Sorkin. Space-time as a causal set. *Physical Review Letters*, 59:521–524, 1987.
- [7] Romeo Brunetti, Klaus Fredenhagen, and Rainer Verch. The generally covariant locality principle—a new paradigm for local quantum field theory. *Communications in Mathematical Physics*, 237:31–68, 2003.
- [8] Paul Busch. Quantum states and generalized observables: A simple proof of gleason’s theorem. *Physical Review Letters*, 91:120403, 2003.
- [9] Giulio Chiribella, Giacomo Mauro D’Ariano, and Paolo Perinotti. Informational derivation of quantum theory. *Physical Review A*, 84:012311, 2011.
- [10] Alain Connes and Carlo Rovelli. Von neumann algebra automorphisms and time-thermodynamics relation in generally covariant quantum theories. *Classical and Quantum Gravity*, 11(12):2899–2917, 1994.
- [11] David Deutsch. Constructor theory. *Synthese*, 190:4331–4359, 2013.
- [12] David Deutsch and Chiara Marletto. Constructor theory of information. *Proceedings of the Royal Society A*, 471:20140540, 2015.
- [13] Bryce S. DeWitt. Quantum theory of gravity. i. the canonical theory. *Physical Review*, 160:1113–1148, 1967.
- [14] Murray Gell-Mann and James B. Hartle. Quantum mechanics in the light of quantum cosmology. In Wojciech H. Zurek, editor, *Complexity, Entropy, and the Physics of Information*, pages 425–458. Addison-Wesley, 1990.

- [15] Andrew M. Gleason. Measures on the closed subspaces of a hilbert space. *Journal of Mathematics and Mechanics*, 6(6):885–893, 1957.
- [16] Henrique Gomes, Sean Gryb, and Tim Koslowski. Einstein gravity as a 3d conformally invariant theory. *arXiv:1010.2481*, 2011.
- [17] Robert B. Griffiths. Consistent histories and the interpretation of quantum mechanics. *Journal of Statistical Physics*, 36:219–272, 1984.
- [18] Lucien Hardy. Quantum theory from five reasonable axioms. *arXiv:quant-ph/0101012*, 2001.
- [19] James B. Hartle. The quantum mechanics of cosmology. In Sidney Coleman, James B. Hartle, Tsvi Piran, and Steven Weinberg, editors, *Quantum Cosmology and Baby Universes*, pages 65–157. World Scientific, 1991.
- [20] Stefan Hollands and Robert M. Wald. Local wick polynomials and time ordered products of quantum fields in curved spacetime. *Communications in Mathematical Physics*, 223:289–326, 2001.
- [21] Stefan Hollands and Robert M. Wald. Existence of local covariant time ordered products of quantum fields in curved spacetime. *Communications in Mathematical Physics*, 231:309–345, 2002.
- [22] Bei-Lok B. Hu and Enric Verdaguer. Stochastic gravity: Theory and applications. *Living Reviews in Relativity*, 11:3, 2008.
- [23] E. T. Jaynes. Information theory and statistical mechanics. *Physical Review*, 106:620–630, 1957.
- [24] E. T. Jaynes. Information theory and statistical mechanics. ii. *Physical Review*, 108:171–190, 1957.
- [25] Leonid Mandelstam and Igor Tamm. The uncertainty relation between energy and time in non-relativistic quantum mechanics. *Journal of Physics (USSR)*, 9:249–254, 1945.
- [26] Ned Markosian and Meghan Sullivan. Time. In Edward N. Zalta and Uri Nodelman, editors, *The Stanford Encyclopedia of Philosophy*. Metaphysics Research Lab, Stanford University, winter 2025 edition, 2025.
- [27] Lluís Masanes and Markus P. Mueller. A derivation of quantum theory from physical requirements. *New Journal of Physics*, 13:063001, 2011.
- [28] Nicholas Maxwell. Are probabilism and special relativity incompatible? *Philosophy of Science*, 52(1):23–43, 1985.
- [29] Hermann Minkowski. Raum und zeit. *Jahresbericht der Deutschen Mathematiker-Vereinigung*, 18:75–88, 1909. Lecture delivered in Cologne, 21 September 1908.
- [30] Don N. Page and William K. Wootters. Evolution without evolution: Dynamics described by stationary observables. *Physical Review D*, 27:2885–2892, 1983.
- [31] Daniel Peterson and Michael Silberstein. Relativity of simultaneity and eternalism: In defense of the block universe. In Vesselin Petkov, editor, *Space, Time, and Spacetime: Physical and Philosophical Implications of Minkowski’s Unification of Space and Time*, pages 209–237. Springer, 2010.

- [32] Hilary Putnam. Time and physical geometry. *The Journal of Philosophy*, 64(8):240–247, 1967.
- [33] C. W. Rietdijk. A rigorous proof of determinism derived from the special theory of relativity. *Philosophy of Science*, 33(4):341–344, 1966.
- [34] Carlo Rovelli. *Quantum Gravity*. Cambridge University Press, 2004.
- [35] Steven Savitt, Natalja Deng, and Oliver Pooley. Being and becoming in modern physics. In Edward N. Zalta, editor, *The Stanford Encyclopedia of Philosophy*. Metaphysics Research Lab, Stanford University, fall 2021 edition, 2021.
- [36] M. M. Taddei, B. M. Escher, L. Davidovich, and R. L. de Matos Filho. Quantum speed limit for physical processes. *Physical Review Letters*, 110:050402, 2013.
- [37] Xinsheng Tan, Dan-Wei Zhang, Zhen Yang, Ji Chu, Yan-Qing Zhu, Danyu Li, Xiaopei Yang, Shuqing Song, Zhikun Han, Zhiyuan Li, Yuqian Dong, Hai-Feng Yu, Hui Yan, Shi-Liang Zhu, and Yang Yu. Experimental measurement of the quantum metric tensor and related topological phase transition with a superconducting qubit. *Physical Review Letters*, 122:210401, 2019.
- [38] Pieter Thyssen. *The Block Universe: A Philosophical Investigation in Four Dimensions*. PhD thesis, KU Leuven, 2020.
- [39] Pieter Thyssen. The rietdijk–putnam–maxwell argument. PhilArchive, 2023.
- [40] Armin Uhlmann. The “transition probability” in the state space of a *-algebra. *Reports on Mathematical Physics*, 9(2):273–279, 1976.
- [41] Robert M. Wald. The back reaction effect in particle creation in curved spacetime. *Communications in Mathematical Physics*, 54:1–19, 1977.
- [42] Min Yu, Dongxiao Li, Jingcheng Wang, Yaoming Chu, Pengcheng Yang, Musang Gong, Nathan Goldman, and Jianming Cai. Experimental estimation of the quantum fisher information from randomized measurements. *Physical Review Research*, 3:043122, 2021.